بسم الله الرحمن الرحيم

People's Democratic Republic of Algeria
Ministry of Higher Education and Scientific Research

Specialty: Computer Security

Advanced Networks

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Semester: 7 Credit: 5 Coefficient: 4

Chapters:

- Introduction to Advanced Computer Networks
- Network Security
- Quality of Service (QoS)
- Emerging Network Technologies
- Advanced Routing and Switching
- Network Management and Monitoring
- Wireless and Mobile Networks
- Case Studies and Practical Applications
- Future Trends and Challenges

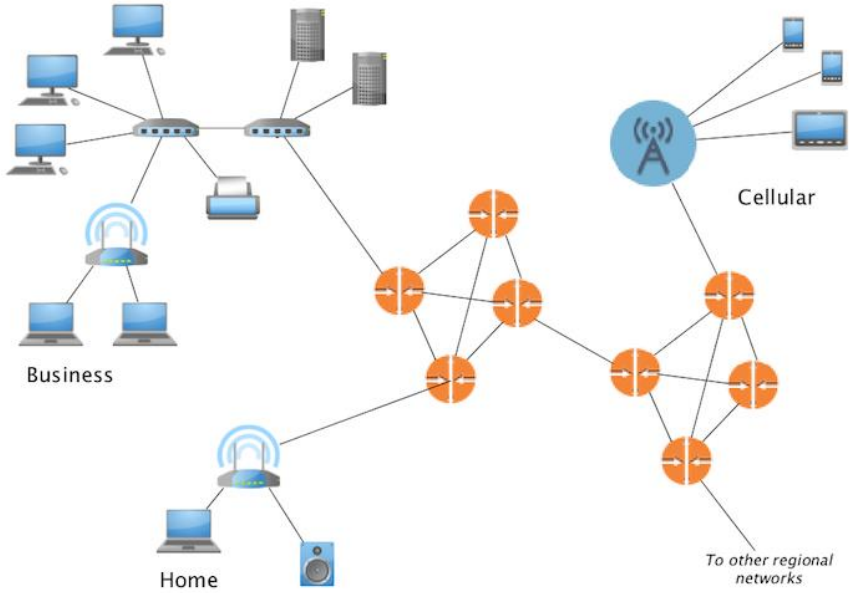
Chapter 01

Introduction to Advanced Computer Networks

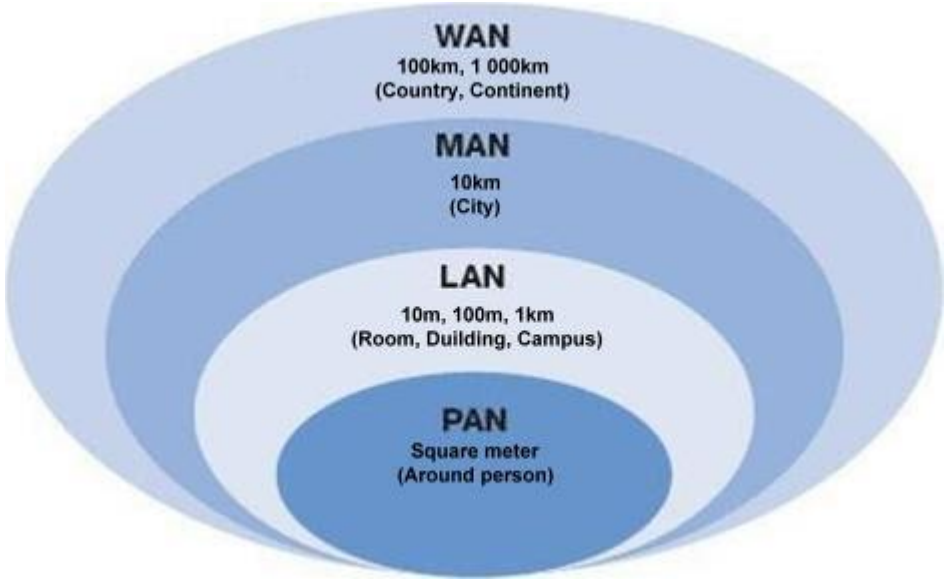
- Objective:
  Overview of the module objectives, structure, and assessment criteria.
  Review of fundamental networking concepts and protocols.
  Introduction to advanced networking topics and their relevance in modern network environments.

What is a Network?

› A computer network is a group of interconnected devices (such as computers, servers, switches, routers, and IoT devices) that communicate with each other to share resources and exchange data. The practice of creating and managing these networks, known as networking, ensures reliable connectivity, security, and performance. Depending on their scope, networks can be classified as Local Area Networks (LAN), Wide Area Networks (WAN), Metropolitan Area Networks (MAN), or wireless networks such as Wi-Fi and cellular systems.



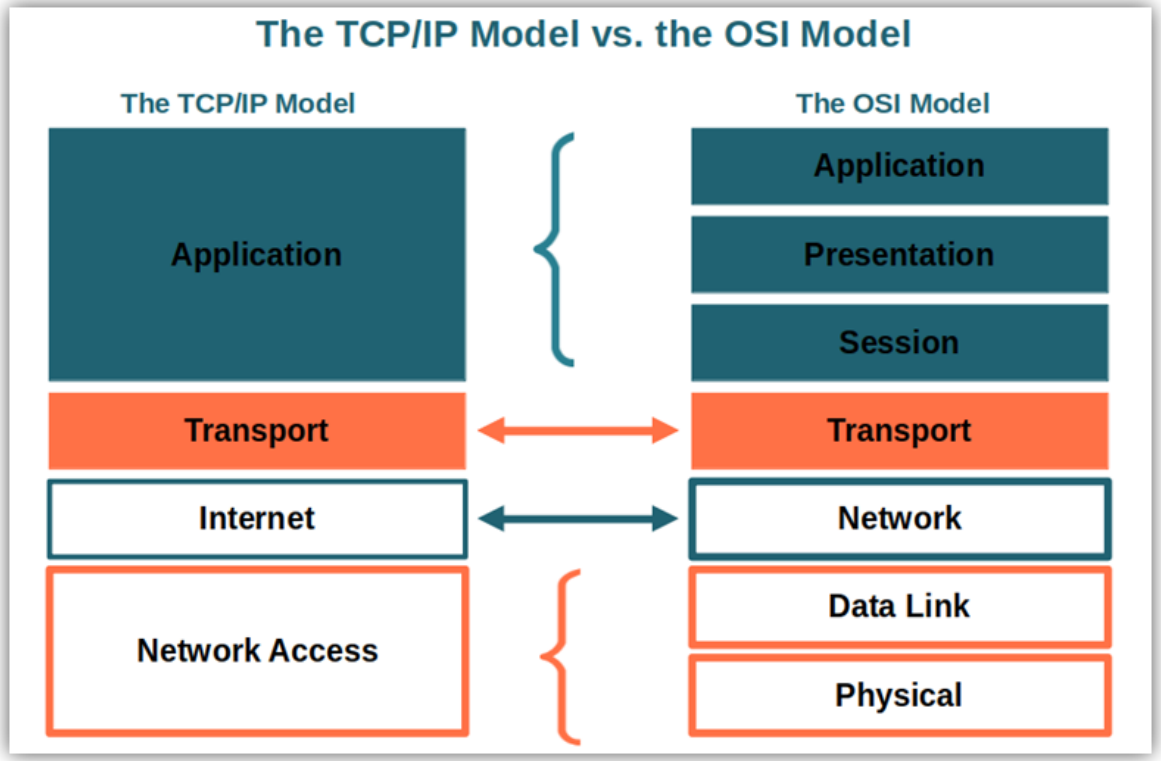
Computer network components



Network classification

Reference Models

To handle network complexity, reference models are used to structure communication into well-defined layers, making it easier to understand, design, and troubleshoot networks. The two main models are the **OSI model**, which organizes communication into seven layers, and the **TCP/IP model**, which simplifies it into four layers. Both models describe how data flows from one device to another, ensuring interoperability between different systems and technologies.



Application Layer: Provides services and interfaces for end-user applications (e.g., HTTP/HTTPS, DNS).
Presentation Layer: Translates, encrypts, and compresses data to ensure proper format between systems (TLS / SSL, ASCII / EBCDIC).
Session Layer: Manages sessions and dialogs between applications, keeping track of connections (NetBIOS).
Transport Layer: Ensures reliable data delivery with error detection, flow control, and segmentation (e.g., TCP, UDP).
Network Layer: Handles logical addressing and routing to deliver packets across different networks (e.g., IP, OSPF).
Data Link Layer: Provides error detection and organizes data into frames for delivery within the same network (e.g., Ethernet, Wi-Fi).
Physical Layer: Transmits raw bits over physical media such as cables, radio waves, or fiber optics (SONET/SDH (fiber optics), RS-232 (serial communication)).

Functions with protocol examples.

Application Protocols

Protocol	Function	How it Works
Hypertext Transfer Protocol / Secure (HTTP / HTTPS)	Enables web communication (client–server).	HTTP sends requests/responses in plaintext; HTTPS wraps HTTP in **TLS** for encryption and authentication.
Domain Name System (DNS)	Resolves domain names to IP addresses.	Uses hierarchical name servers (root → TLD → authoritative). Queries usually over UDP/53, zone transfers over TCP.
Transport Layer Security (TLS)	Secures communication with encryption & authentication.	Uses certificates (PKI) and symmetric/asymmetric cryptography to establish a secure session (handshake + data encryption).
American Standard Code for Information Interchange (ASCII coding)	Represents characters as numbers (0–127).	Maps each letter/symbol to a numeric binary code (e.g., “A” = 65). Ensures consistent text exchange across systems.
Network Basic Input/Output System (NetBIOS)	Provides legacy naming and session services in LANs.	Assigns names to devices and manages sessions over TCP/UDP ports (137–139). Replaced mostly by DNS.

Transport & Network Protocols

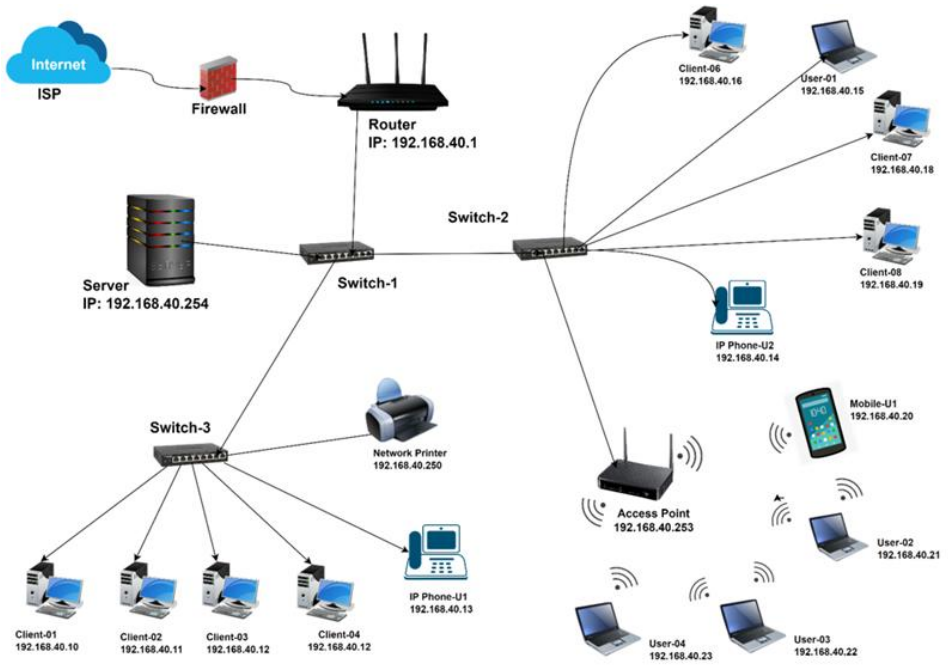
Protocol	Function	How it Works
Transmission Control Protocol (TCP)	Reliable, connection-oriented transport.	Uses 3-way handshake, acknowledgments, sequencing, retransmission, and flow control to guarantee delivery.
User Datagram Protocol (UDP)	Fast, connectionless transport.	Sends datagrams without setup, no retransmission or ordering. Used in real-time apps (VoIP, video, gaming).
Internet Protocol (IP v4/v6)	Provides logical addressing and routing.	Encapsulates data into packets with source/destination IP. Routers forward based on routing tables.
Open Shortest Path First (OSPF)	Dynamic routing protocol (link-state).	Builds a complete topology map, uses Dijkstra's algorithm to compute shortest paths. Organizes networks into hierarchical areas.

Network Devices

- **Hub**
  A hub was an early device used to connect multiple devices in a LAN.
  Unlike switches, hubs simply **broadcast** all incoming data to every port.
  They are rarely used today because they are inefficient and insecure.

- **Switch**
  A switch connects multiple devices (computers, servers, printers) within a Local Area Network (LAN).
  It operates by learning and using MAC addresses to forward frames only to the intended recipient, unlike hubs that broadcast to all devices.
  Modern switches also support advanced features like VLANs, link aggregation, and sometimes even routing.

- **Router**
  A router connects different networks together, such as your home network to the Internet.
  It makes forwarding decisions based on IP addresses and relies on routing protocols like OSPF or BGP to determine the best path for data.
  Routers can also provide NAT (Network Address Translation), DHCP services, and basic firewalling.



Network Devices

- **Access Point (AP)**

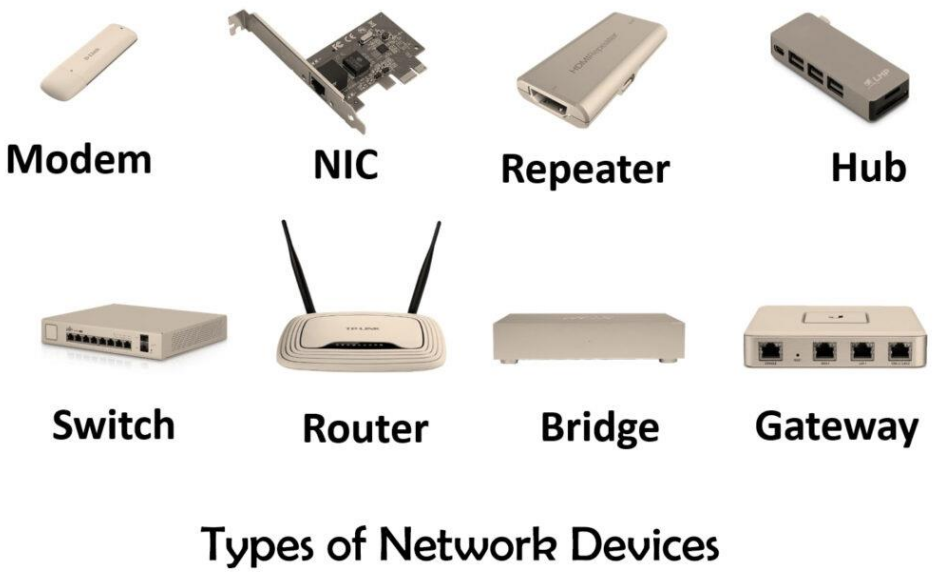
An access point provides **wireless connectivity**, allowing devices like laptops and smartphones to join a network without cables.
It acts as a bridge between Wi-Fi clients and the wired LAN.
Enterprise APs often include features such as user authentication, roaming support, and centralized management.

- **Firewall**

A firewall protects a network by controlling the traffic that enters or leaves it.
It enforces security policies by allowing or blocking packets based on rules.
Types include **packet-filtering firewalls**, **stateful firewalls**, and **next-generation firewalls (NGFW)** with intrusion detection and deep packet inspection.

- **Gateway**

A gateway connects networks that use **different protocols or architectures**.
It translates data between systems (e.g., a VoIP gateway that converts phone calls between IP and traditional phone networks).
Often used in **IoT** and **industrial networks** for interoperability.



What is Advanced Networking?

Advanced Networking refers to the study and implementation of **modern and sophisticated networking technologies, protocols, and architectures** that go beyond basic connectivity.
While traditional networking focuses on **establishing communication** between devices (using IP, routing, and switching), **advanced networking** emphasizes **optimizing, securing, and automating** these communications to meet the complex demands of today's digital environments.
It involves technologies and concepts such as:

- **Network Security** (firewalls, IDS/IPS, encryption)
- **Quality of Service (QoS)** for performance assurance
- **Software-Defined Networking (SDN)** and **Network Function Virtualization (NFV)**
- **Wireless and Mobile Networks (4G/5G, IoT)**
- **Advanced routing and switching** (BGP, OSPF, VLANs)
- **Network management and monitoring tools** (SNMP, NetFlow)

Why Advanced Networking

Increased Complexity of Modern Networks

Today's networks integrate wired, wireless, cloud, and virtualized components.

Managing and securing such environments requires advanced protocols and automation tools.

Security Challenges

As cyber threats evolve, network professionals must understand advanced mechanisms (e.g., encryption, VPNs, IDS/IPS) to protect data in transit.

Performance and QoS Requirements

Applications like video conferencing, online gaming, and telemedicine demand **low latency**, **high bandwidth**, and **reliable performance**, all managed via advanced QoS techniques.

Scalability and Virtualization

Cloud data centers and virtual networks (via SDN/NFV) enable dynamic resource allocation, essential for large-scale systems and enterprise operations.

Support for Emerging Technologies

Advanced networks are the **backbone of 5G, IoT, AI-driven networking, and edge computing**, all of which rely on flexible, intelligent network infrastructures.

Automation and Management Efficiency

Advanced networking introduces **programmable networks** (e.g., SDN) that simplify configuration, reduce human error, and increase adaptability.

Chapter 02

Network Security

- Content:
  * Threats, vulnerabilities, and attacks in computer networks.
  * Cryptography and encryption techniques for securing data transmission.
  * Firewalls, intrusion detection systems, and other security mechanisms.
  * Secure network design principles and best practices.

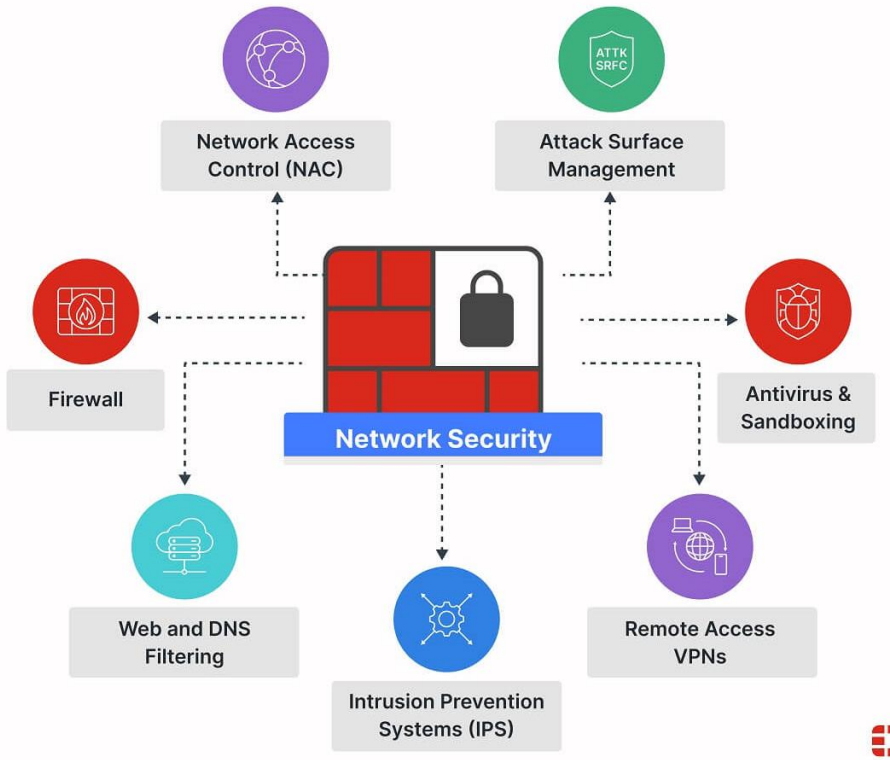
What Is Network Security?

It refers to the **policies, technologies, and practices** designed to protect data and resources from unauthorized access, misuse, or attacks.

It involves both **hardware and software solutions**, including firewalls, intrusion prevention systems, and secure protocols.

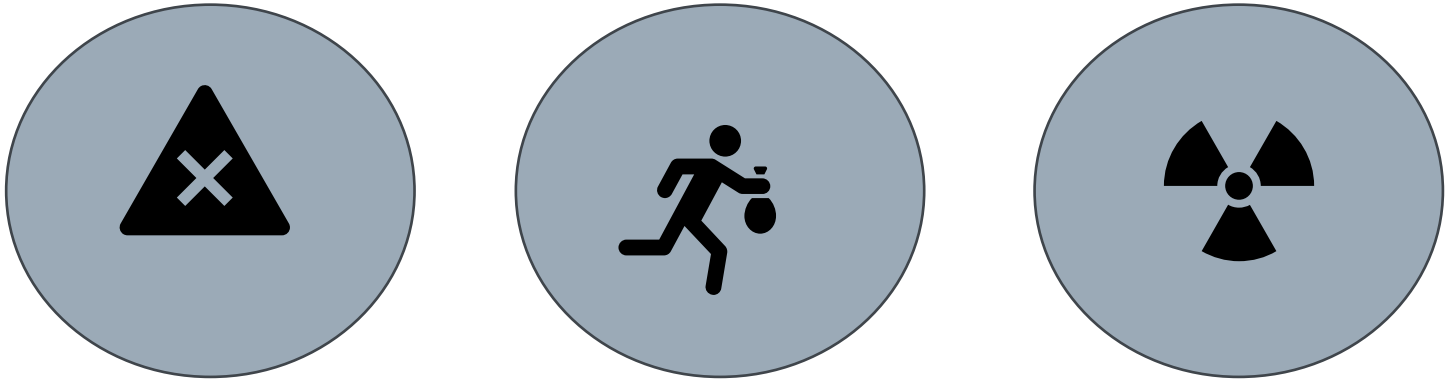
Core principles include:

- **Confidentiality**: Only authorized users can access data.
- **Integrity:** Data remains accurate and unchanged.
- **Availability**: Systems and information are accessible when needed.
- **Authentication & Authorization**: Verify identity and control access.
- **Accountability**: Actions can be traced and audited.



Threats, Vulnerabilities, and Attacks

- A **threat** is any potential event, action, or circumstance that can cause harm to a network, system, or organization by exploiting a vulnerability. In networking, threats may target confidentiality (data theft), integrity (data modification), or availability (service disruption).
- A **vulnerability** is a weakness, flaw, or gap in the design, implementation, or configuration of a system that could be exploited by a threat. Vulnerabilities can exist in hardware, software, network configurations, or human behavior (e.g., poor password practices).
- An **attack (risk)** is the deliberate action taken by a threat actor to exploit a vulnerability and compromise the target system's security. Attacks can aim to steal data, disrupt services, or gain unauthorized control over systems.



Common Network Threats, Attacks, and Vulnerabilities

Type	Category	Description	Examples
Threat	Internal Threats	Originate inside the organization; may result from intentional misuse or human error.	Malicious insider, careless employee, data leakage.
	External Threats	Come from outside sources attempting to compromise or disrupt the network.	Hackers, competitors, cybercriminals
	Environmental Threats	Non-human factors that threaten system availability.	Power outage, hardware failure, fire, flood.
Vulnerability	System & Configuration Weaknesses	Flaws in design, configuration, or management that can be exploited.	Unpatched OS, weak passwords, open ports, default credentials.
	Protocol Weaknesses	Insecure or outdated communication standards.	FTP, Telnet, HTTP (unencrypted protocols).
	Physical & Environmental Weaknesses	Inadequate physical protections or environmental safeguards.	Unsecured server rooms, lack of backup power, overheating.

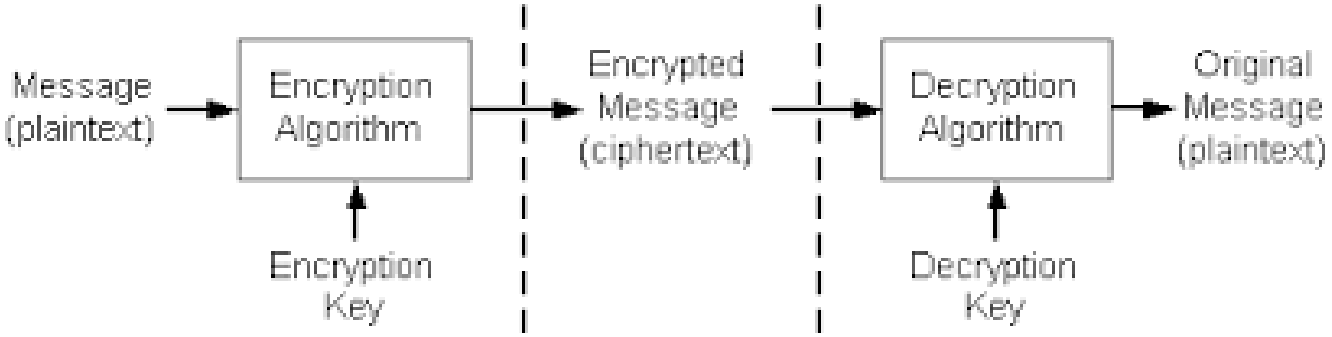
Common Network Threats, Attacks, and Vulnerabilities

Type	Category	Description	Examples
Attack	Active Attacks	Attempt to alter or disrupt network operations and data integrity.	DoS/DDoS, spoofing, man-in-the-middle (MITM), session hijacking.
	Passive Attacks	Aim to monitor, capture, or analyze network traffic without changing it.	Packet sniffing, eavesdropping, traffic analysis.
	Network Attacks	Target the infrastructure or protocols to gain control or cause disruption.	IP/MAC spoofing, port scanning, routing attacks (e.g., OSPF poisoning).
	Malware-Based Attacks	Use malicious software to damage, steal, or control data/systems.	Viruses, worms, Trojans, ransomware, spyware.
	Human-Centric Attacks	Exploit human trust or behavior rather than technical flaws.	Phishing, social engineering, shoulder surfing.

WHAT IS CRYPTOGRAPHY?

Cryptography is the science and art of securing information by transforming it into a form that is unreadable to unauthorized users. It ensures that even if data is intercepted or accessed by an attacker, it cannot be understood or altered without the proper cryptographic keys.

- **Key Concepts**
- **Plaintext:** The original readable message or data.
- **Ciphertext:** The encrypted, unreadable form of the plaintext.
- **Key:** A secret value used by an algorithm to encrypt or decrypt data.
- **Encryption:** The process of converting plaintext into ciphertext.
- **Decryption:** The process of converting ciphertext back into readable plaintext.
- **Algorithm (Cipher):** A mathematical function that performs encryption or decryption using a key.



CRYPTOGRAPHIC TECHNIQUES

Symmetric Encryption

Asymmetric Encryption

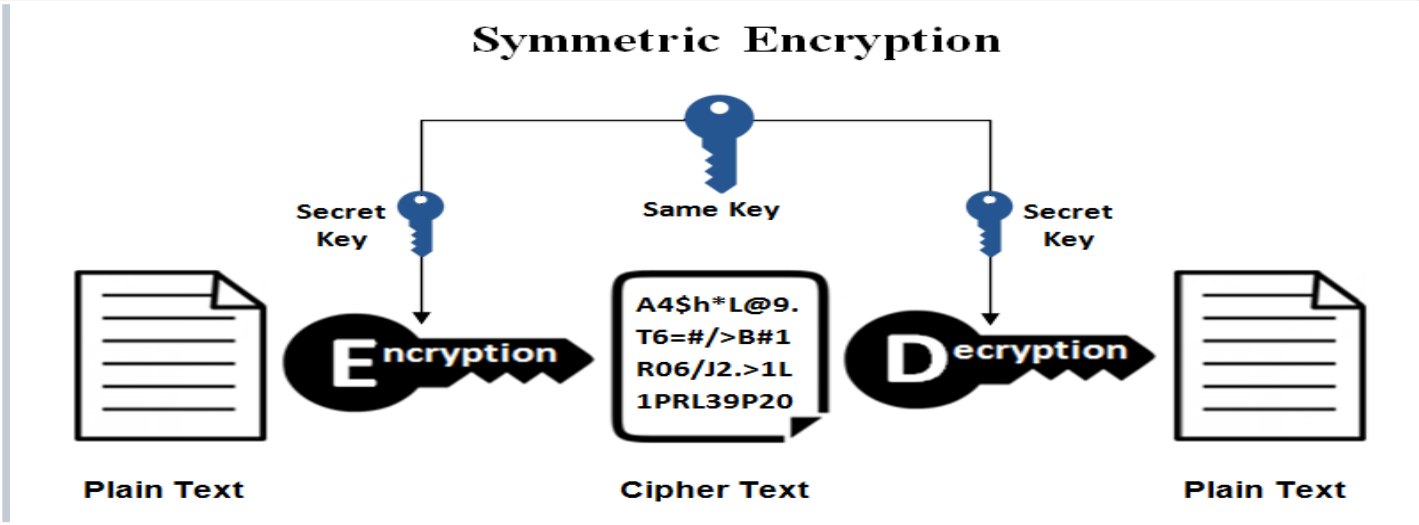
Hashing

Digital Signatures

Hybrid Encryption

Symmetric encryption (secret-key encryption like AES, DES, and chacha20) is a cryptographic technique where the same key is used for both encryption and decryption. It is the fastest and most efficient form of encryption, commonly used for securing large volumes of data in modern networks.

1. The sender and receiver share a single secret key in advance.
2. The sender uses this key and an encryption algorithm to convert plaintext into ciphertext.
3. The receiver uses the same key and a decryption algorithm to recover the original plaintext.



Advantages and Disadvantages of Symmetric Encryption

Advantages:
•High speed: Efficient for encrypting large data volumes.
•Low computational cost: Ideal for hardware or software implementation.
•Strong security (with modern algorithms like AES).

Disadvantages
•Key distribution problem: Both parties must securely share the same key.
•Scalability issue: For n users, you need $n(n-1)/2$ unique keys.
•No built-in authentication: It only ensures confidentiality — not identity verification.

CRYPTOGRAPHIC TECHNIQUES

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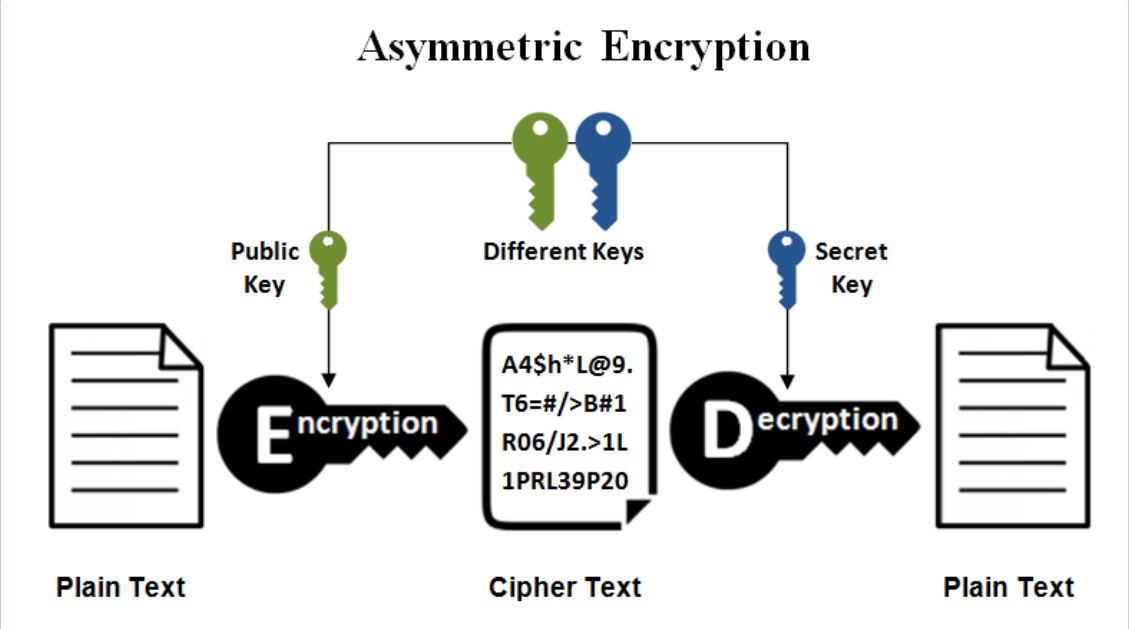
Hybrid Encryption

Asymmetric encryption, like RSA, is a cryptographic technique that uses two different keys (a public key and a private key) for secure communication. The public key is shared openly and used for encryption. The private key is kept secret and used for decryption. The two keys are mathematically related, but it's computationally infeasible to derive one from the other.

1. Each user generates a key pair (public + private).
2. The public key is distributed to others.
3. When someone wants to send you a secure message:
   They encrypt it using your public key.
   You decrypt it using your private key.
4. Only you private key can decrypt data encrypted with your public key.



Advantages and Disadvantages of Symmetric Encryption

Advantages

•Solves the key distribution problem (no need to share secret keys beforehand).
•Enables authentication and non-repudiation (when combined with digital signatures).
•Supports secure key exchange for symmetric encryption.

Disadvantages

•Slower than symmetric encryption due to complex mathematics.
•Not suitable for large data encryption — typically used only to exchange symmetric keys.
•Relies on trust infrastructure (e.g., certificates, PKI).
•Vulnerable to private key compromise — if the private key is stolen, security is lost.

CRYPTOGRAPHIC TECHNIQUES

Symmetric Encryption

Asymmetric Encryption

Hashing

Digital Signatures

Hybrid Encryption

Hashing is a mathematical process that transforms an input message of *arbitrary length* (text, file, password, etc.) into a fixed-size string of characters called a hash value, digest, or message digest, using a hash function.

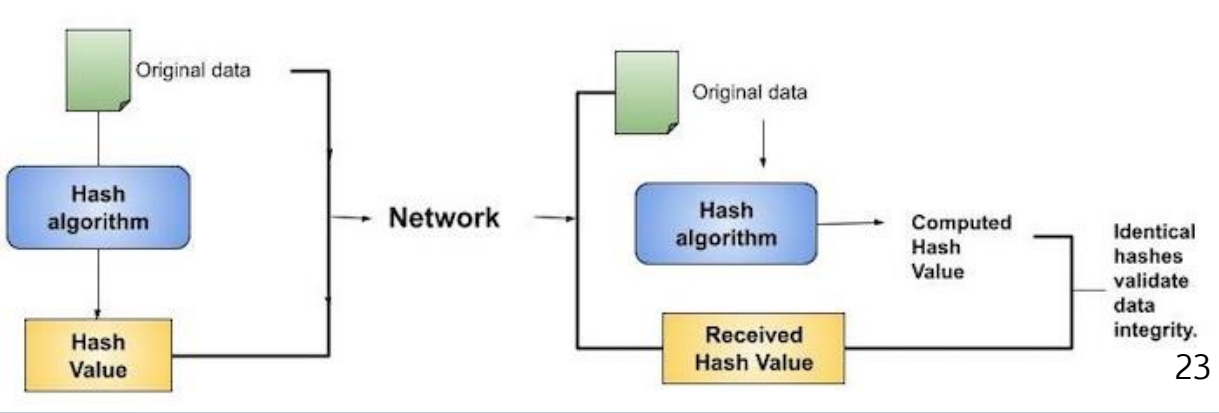
It is a one-way and deterministic operation:

One-way means it is computationally infeasible to reverse the process — you cannot reconstruct the original input from its hash value.

Deterministic means the same input always produces the same hash output.

The hash function performs a series of logical, bitwise, and modular arithmetic operations on blocks of input data, generating a fixed-size output (for example, 256 bits for SHA-256), regardless of how large or small the input is.

"Hashing does not provide confidentiality, but it ensures integrity (meaning that even the smallest change in input data (like changing one bit) results in a completely different hash value."



Why Hashing Is Still a Type of Security (Even Without Confidentiality)

Hashing does not hide information (so it doesn't provide *confidentiality*),
but it protects the trustworthiness and authenticity of data, which are *other* pillars of security.
In cybersecurity, we often talk about the CIA triad (Confidentiality, Integrity, and Availability).
Hashing directly supports the I: Integrity part.

Integrity as a Security Goal:

Integrity ensures that data has not been changed, deleted, or tampered with, whether accidentally (e.g., transmission error) or intentionally (e.g., attacker modification).
If integrity is broken:

- A message, file, or transaction could be altered.
- Even encrypted data could become invalid or dangerous.

Hashing detects such modifications instantly. That's why it's a core security mechanism, even without confidentiality.

CRYPTOGRAPHIC TECHNIQUES

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Digital Signatures

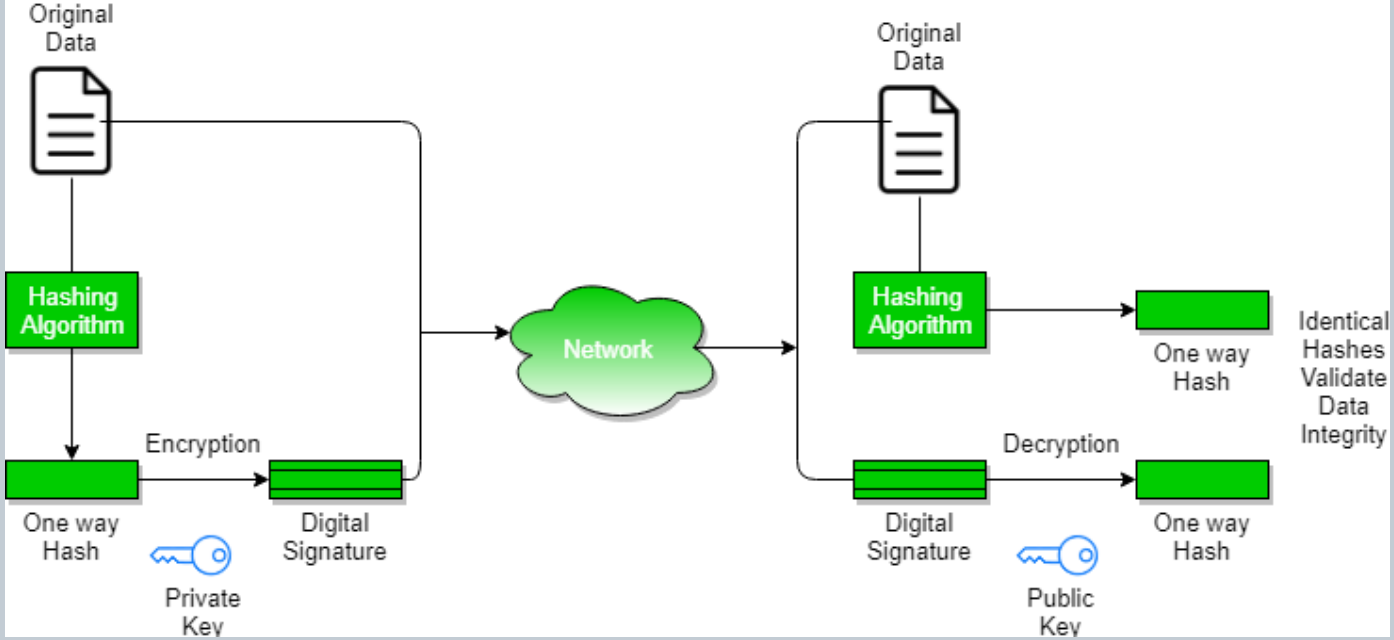
Hybrid Encryption

A digital signature is a cryptographic mechanism that provides authentication, integrity, and non-repudiation for digital data.

It works like an electronic version of a handwritten signature ensuring that:

- The message really came from the claimed sender (*authentication*),
- The message has not been altered (*integrity*), and
- The sender cannot deny having sent it (*non-repudiation*).

Digital signatures are based on asymmetric cryptography and hashing.



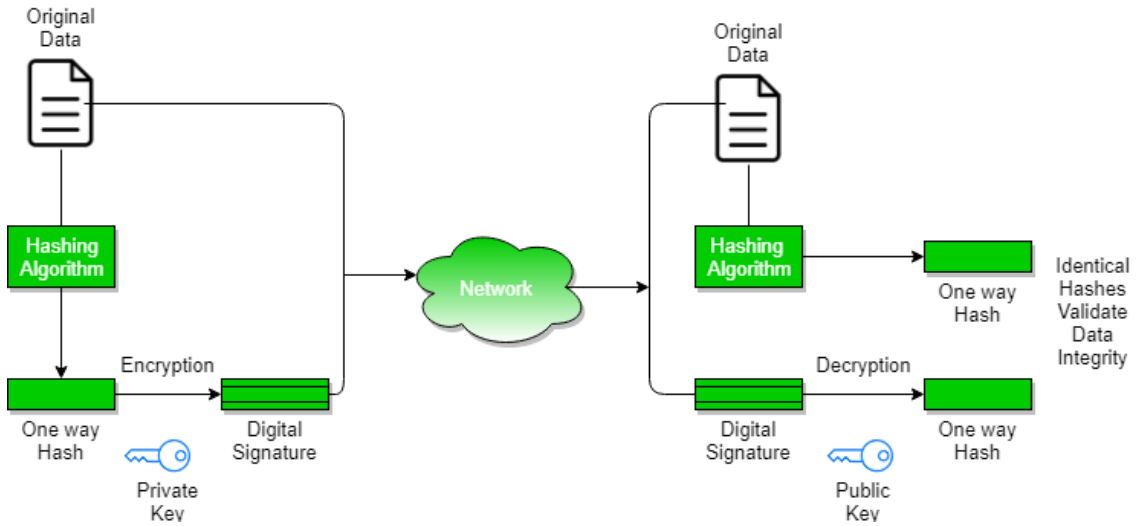
Digital Signature Step by Step

Signing Process (Sender's Side)

1. The sender computes a **hash** of the message (e.g., SHA-256).
2. The hash (not the message itself) is **encrypted using the sender's private key**.
3. This encrypted hash is the **digital signature**.
4. The signature is sent along with the message.

2. Verification Process (Receiver's Side)

1. The receiver computes a **hash** of the received message.
2. The receiver **decrypts the digital signature** using the sender's **public key**, obtaining the sender's original hash.
3. The receiver **compares** the two hashes:
    1. If they match → The message is authentic and unchanged.
    2. If they differ → The message was altered or not from the claimed sender.



CRYPTOGRAPHIC TECHNIQUES

Symmetric Encryption

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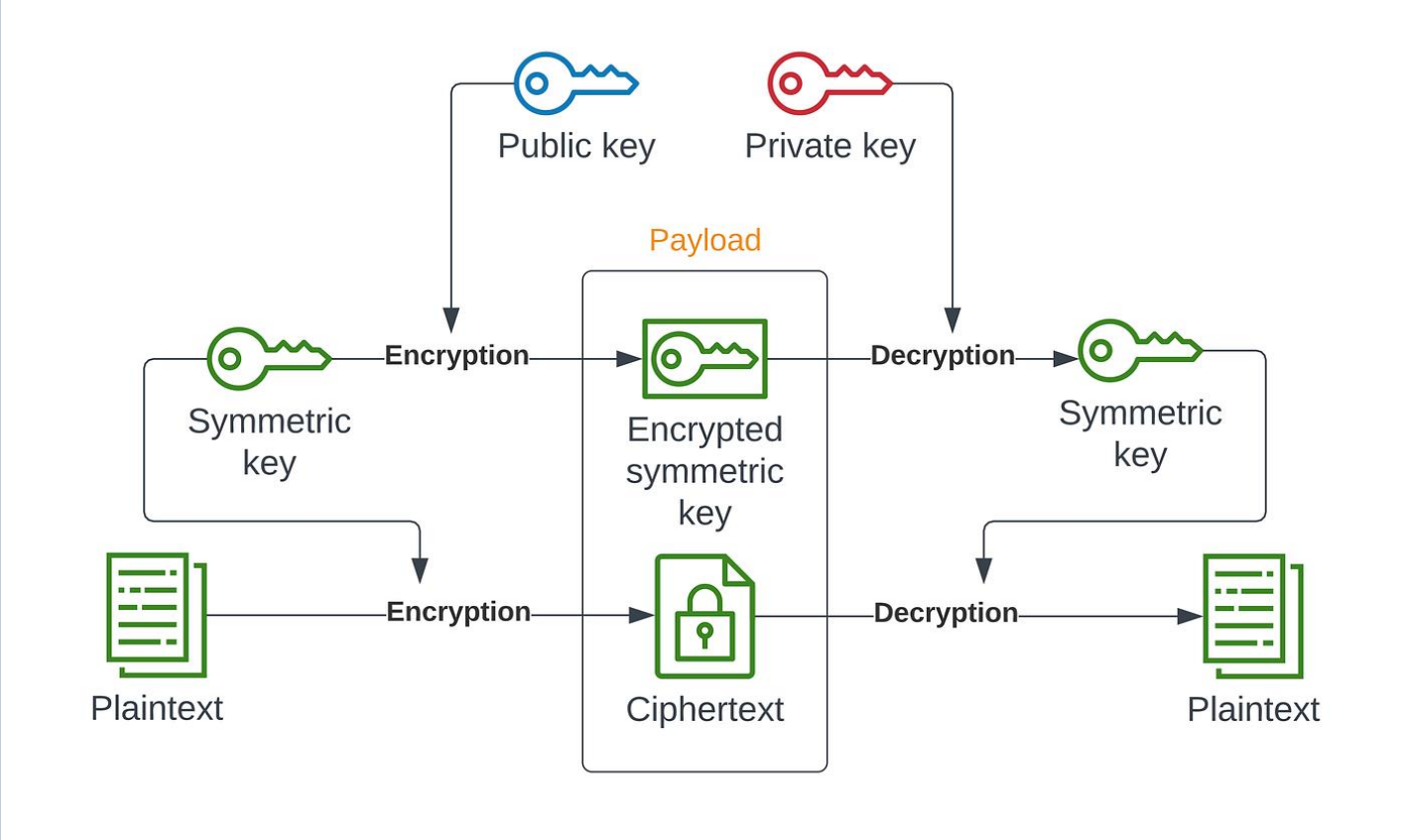
Digital Signatures

Hybrid Encryption

Hybrid encryption is a cryptographic approach that combines symmetric and asymmetric encryption to take advantage of the strengths of both techniques, the speed and efficiency of symmetric encryption and the secure key exchange of asymmetric encryption.



Hybrid Encryption Step by Step

1.Key Generation:

1. The sender generates a random symmetric key (also called a *session key*).
2. This key will be used to encrypt the actual data using a fast symmetric algorithm (e.g., AES).

2.Data Encryption:

1. The plaintext data (file, message, etc.) is encrypted with the symmetric key, producing ciphertext.
2. This ensures fast and efficient data encryption, even for large files.

3.Key Encryption:

1. The symmetric key itself is then encrypted using the receiver's public key (asymmetric encryption, e.g., RSA).
2. Only the receiver can decrypt this key using their private key, ensuring secure key exchange.

4.Transmission:

1. The sender transmits both:
    1. The encrypted symmetric key, and
    2. The encrypted data (ciphertext) to the receiver.

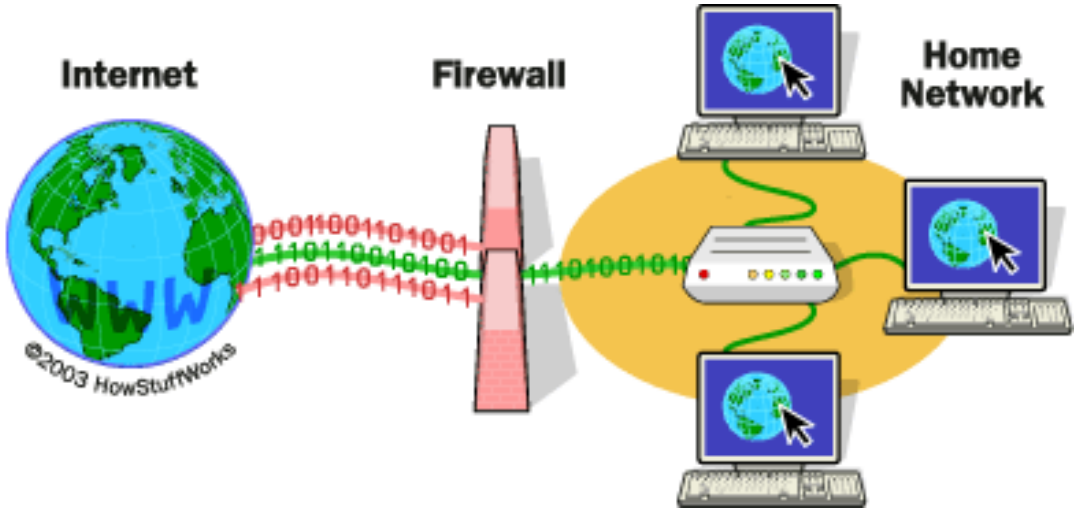
5.Decryption (Receiver Side):

1. The receiver first decrypts the symmetric key using their private key.
2. Then, the decrypted symmetric key is used to decrypt the data back into plaintext.

Firewall and its Functionality

A firewall is a network security device or software that monitors and controls incoming and outgoing traffic based on predefined security rules.
It acts as a barrier between trusted internal networks and untrusted external networks (like the Internet).
Primary Functionality:

- To manage traffic flow by permitting or denying packets based on criteria such as IP addresses, port numbers, and transport protocols.
- To implement security policies that safeguard the network against unauthorized access attempts.



FIREWALL TYPES

Packet-Filtering Firewall

Stateful Inspection Firewall

Application-Level Firewall (Proxy Firewall)

Next-Generation Firewall (NGFW)

It is a basic firewall operates primarily at Layer 3 (Network Layer) of the OSI model, analyzing packet headers to enforce security policies. It focuses on attributes such as source and destination IP addresses, port numbers, and protocol types.

Operational Mechanism:

- Upon transmission, each packet undergoes scrutiny against a predefined set of rules encapsulated in an Access Control List (ACL).
- If a packet adheres to an 'allow' rule, it is permitted to proceed; if not, the packet is discarded.

Benefits:

- The firewall's simplicity ensures rapid processing.
- Its efficiency yields low overhead on system resources.

Limitations:

- The firewall operates solely on header information, lacking the capacity to perform deep packet inspection.
- It cannot maintain stateful connection tracking, which poses risks of permitting spoofed or fragmented packets.
- As a stateless mechanism, it offers no defense against application-layer threats such as malware or SQL injection attacks.

FIREWALL TYPES

Packet-Filtering Firewall

Stateful Inspection Firewall

Application-Level Firewall (Proxy Firewall)

Next-Generation Firewall (NGFW)

Operation Principle

A stateful firewall operates by maintaining a state table to monitor active connections, ensuring that only packets belonging to established sessions are permitted through.

Mechanism

Upon initiation of a connection, such as a TCP session, the firewall logs critical session parameters, including IP addresses, port numbers, and sequence numbers. It employs this information to verify the legitimacy of incoming packets against known sessions. This methodology effectively blocks unauthorized packets that attempt to masquerade as legitimate traffic.

Benefits

Stateful firewalls offer enhanced security over stateless packet filtering mechanisms by providing context for each packet. They are capable of recognizing and intercepting anomalous TCP handshake processes, further strengthening network defenses.

Limitations

However, they incur slightly increased resource overhead due to the necessity of state tracking. Additionally, stateful firewalls do not offer the capability to inspect application-layer data, which can limit their effectiveness against certain types of threats.

FIREWALL TYPES

Packet-Filtering Firewall

Stateful Inspection Firewall

Application-Level Firewall (Proxy Firewall)

Next-Generation Firewall (NGFW)

A proxy firewall functions as an intermediary layer between clients and external networks, facilitating controlled communication. Rather than allowing direct traffic flow, it intercepts client requests, thoroughly inspects them, and relays them to the intended destinations.

Operational Overview

1. Client Connection: The client establishes a connection to the proxy firewall (e.g., an HTTP proxy).
2. Content Examination: The proxy conducts a detailed analysis of the incoming requests, evaluating components such as URLs and commands for potential threats.
3. New Connection Initiation: Upon validating the safety of the requests, the proxy initiates a fresh connection to the target server, effectively representing the client's request.

Benefits

- Enables in-depth traffic content inspection, providing enhanced security and compliance capabilities.
- Capable of caching responses, which significantly improves response times for repeated requests.
- Masks internal IP addresses, providing a layer of anonymity and simulating a form of Network Address Translation (NAT).

Limitations

- Content inspection can introduce additional latency, potentially impacting user experience.
- Each protocol (e.g., HTTP, FTP, etc.) requires a dedicated proxy solution, complicating deployment and management.

FIREWALL TYPES

Packet-Filtering Firewall

Stateful Inspection Firewall

Application-Level Firewall (Proxy Firewall)

Next-Generation Firewall (NGFW)

Next-Generation Firewalls (NGFW) extend the capabilities of traditional firewalls by incorporating advanced security functions, including:
- Deep Packet Inspection (DPI): Enables thorough analysis of packet data beyond header information.
- Intrusion Prevention Systems (IPS): Actively identifies and mitigates known vulnerabilities and threats in real-time.
- Application Awareness and Control: Grants the ability to discern and manage application-specific traffic irrespective of the transmitting port.
- SSL/TLS Decryption: Facilitates inspection of encrypted traffic to ensure malicious content isn't overlooked.
- Malware Detection and Sandboxing**: Detects and analyzes potentially harmful code in a safe environment before it can impact the network.

Operational Mechanism:
NGFWs analyze traffic at the application layer, employing sophisticated techniques that identify and manage specific applications (such as Skype or YouTube) irrespective of their respective ports. This enables the firewall to apply customized policies that either block or prioritize traffic based on content, user identity, or organizational policies.

FIREWALL TYPES

Packet-Filtering Firewall

Stateful Inspection Firewall

Application-Level Firewall (Proxy Firewall)

Next-Generation Firewall (NGFW)

Benefits

- Provides comprehensive protection against contemporary cyber threats.
- Integrates a suite of security features into a single platform, streamlining management.
- Highly suitable for enterprise-level networks that require robust security postures.

Limitations

- Typically entails higher costs and demands more resources for deployment.
- Configuration and ongoing maintenance can be complex, necessitating skilled personnel.

Firewall types

Firewalls have evolved from simple packet filters to intelligent, application-aware systems. Modern networks rely on **stateful**, **proxy**, and **next-generation firewalls** to provide layered protection across multiple OSI layers.

Type	Layer (OSI)	Main Function	Key Advantage	Main Limitation
Packet Filtering	3	Checks IP, port, protocol	Simple, fast	No payload inspection
Stateful Inspection	3–4	Tracks session states	More secure	Limited to transport layer
Circuit-Level Gateway	5	Verifies session establishment	Validates sessions	No deep inspection
Proxy Firewall	7	Filters application data	Deep inspection	Slower, per-protocol
NGFW	3–7	DPI + IPS + App Control	All-in-one protection	Expensive, complex

Intrusion Detection System (IDS)

Definition:
An Intrusion Detection System (IDS) is a comprehensive security framework designed to monitor and analyze both network and system activities in order to identify suspicious behavior, security breaches, or confirmed attack patterns. The primary purpose of an IDS is to enhance the security posture of an organization by promptly detecting potential threats.

Functionality:
-Detection of Unauthorized Access: The IDS continuously scans for attempts to gain unauthorized entry into systems, ensuring that any illicit access is flagged immediately.
- Policy Violation Alerts: The system evaluates activities against established security policies, alerting administrators when violations occur, such as improper access to sensitive data or unauthorized changes to configurations.
- Anomaly Detection: By monitoring system performance metrics, the IDS can identify deviations from normal operation, signaling potential intrusions or system malfunctions.
- Alert Mechanism: When the IDS identifies potential threats, it generates alerts for system administrators, providing them with timely notifications to investigate and respond to incidents.
- Non-Intrusive Monitoring: Unlike Intrusion Prevention Systems (IPS), IDS does not take active measures to block or prevent traffic; instead, it operates in a passive mode, focusing solely on monitoring, logging, and reporting.

IDS Types and Detection Techniques

IDS Types:

- **Network-based IDS (NIDS):** This type of system monitors data traffic flowing across the network infrastructure. NIDS can analyze packets, inspect communication patterns, and detect potential threats by assessing the entirety of the network's traffic in real-time.
- **Host-based IDS (HIDS):** HIDS focuses on monitoring the activities and events occurring on a specific host or server. This includes tracking file integrity, system calls, and processes, thereby providing a granular level of monitoring for critical servers and endpoints.

Detection Techniques:

- **Signature-based Detection:** This approach involves comparing network traffic against a database of known attack signatures or patterns. Similar to antivirus software, it relies on established threat definitions to identify malicious traffic. While effective for known threats, it may miss newly developed or sophisticated attacks that do not match existing signatures.
- **Anomaly-based Detection:** Utilizing advanced statistical models, algorithms, or machine learning techniques, this method establishes a baseline of normal behavior for systems or networks. It then detects unusual activity that deviates from this baseline, allowing for the identification of previously unknown threats and zero-day vulnerabilities.

Intrusion Prevention System (IPS)

Definition:
An Intrusion Prevention System (IPS) is an advanced security mechanism designed to enhance network defense by not only detecting potential threats but also proactively blocking or mitigating them in real-time. Unlike an Intrusion Detection System (IDS), which solely monitors and alerts on suspicious activity without taking action, an IPS takes immediate countermeasures to safeguard network integrity.

Function:
- Real-Time Network Monitoring: An IPS continuously analyzes network traffic streams to identify patterns that may suggest malicious intent, allowing for rapid detection of anomalies.
- Packet Inspection: It utilizes deep packet inspection (DPI) techniques to scrutinize the contents of data packets, discerning between benign and harmful traffic. Malicious packets can be identified based on predefined signatures or behavioral analysis.
- Automated Response: Upon detecting an attack, the IPS can take predefined actions such as dropping malicious packets, resetting connections, or even blocking the offending IP addresses to prevent further incidents.
- Integration with Next-Generation Firewalls (NGFW): Often, IPS functionality is embedded within Next-Generation Firewalls, providing a more cohesive and comprehensive security solution. This integration allows for more sophisticated traffic filtering and threat detection, combining the features of traditional firewalls with advanced malware detection.

Other Network Defense Mechanisms

1.Network Access Control (NAC):
Ensures only authenticated and compliant devices can connect to the network.
2.Virtual Private Network (VPN):
Encrypts communication between remote users and the internal network, ensuring confidentiality and integrity over public links.
3.Demilitarized Zone (DMZ):
A separate subnetwork that isolates public-facing services (like web servers) from the internal LAN — limits exposure if a server is compromised.
4.Antivirus and Anti-Malware Gateways:
Inspect files and emails for malicious content before allowing them into the network.
5.Security Information and Event Management (SIEM):
Collects and analyzes logs from multiple security systems to detect complex or correlated attacks.

In a secure network, multiple defense mechanisms work together in a **layered approach**. **Firewalls** filter traffic and block unauthorized access, while **IDS/IPS** detect and prevent intrusions in real time.
NAC and **VPNs** ensure secure and trusted connections, and a **DMZ** isolates public services from the internal network. Finally, **SIEM** systems centralize monitoring and alerting, providing complete visibility and faster incident response.

Secure Network Design Principles - Best Practices -

Defense in Depth: Multiple layers of protection.
Least Privilege: Limit access rights.
Segmentation: Isolate sensitive zones (e.g., DMZ, VLANs).
Strong Access Control: MFA, RBAC, NAC.
Encryption: Protect data in transit and at rest.
System Hardening: Disable unused ports/services.
Monitoring: Log and analyze all critical activities.
Patch Management: Keep systems up to date.
Resilience: Redundant paths and failover mechanisms.
Security by Design: Integrate protection from the start.

Secure Network Design Principles -Best Practices-

Principle	Description
Defense in Depth	Use multiple layers of security to protect the network even if one layer fails.
Least Privilege	Grant users and systems only the access rights necessary to perform their specific tasks.
Segmentation and Isolation	Divide the network into smaller subnets or VLANs to contain traffic and limit the spread of attacks.
Secure Access Control	Enforce strong authentication (e.g., MFA) and authorization (e.g., RBAC, NAC) to manage who can access what.
Encryption and Data Protection	Encrypt data both in transit (TLS, VPN) and at rest (AES) to maintain confidentiality and integrity.
Secure Configuration and Hardening	Disable unused services, close unnecessary ports, and replace default credentials to reduce exposure.
Monitoring and Logging	Continuously monitor traffic and log critical events using SIEM systems for detection and forensic analysis.
Regular Updates and Patch Management	Keep software, firmware, and systems up to date to close vulnerabilities before they are exploited.
Redundancy and Resilience	Use backup links, redundant hardware, and failover systems to ensure availability and reliability.
Security by Design	Integrate security considerations from the start of network planning and throughout its lifecycle.

Chapter 03

Quality of Service (QoS)

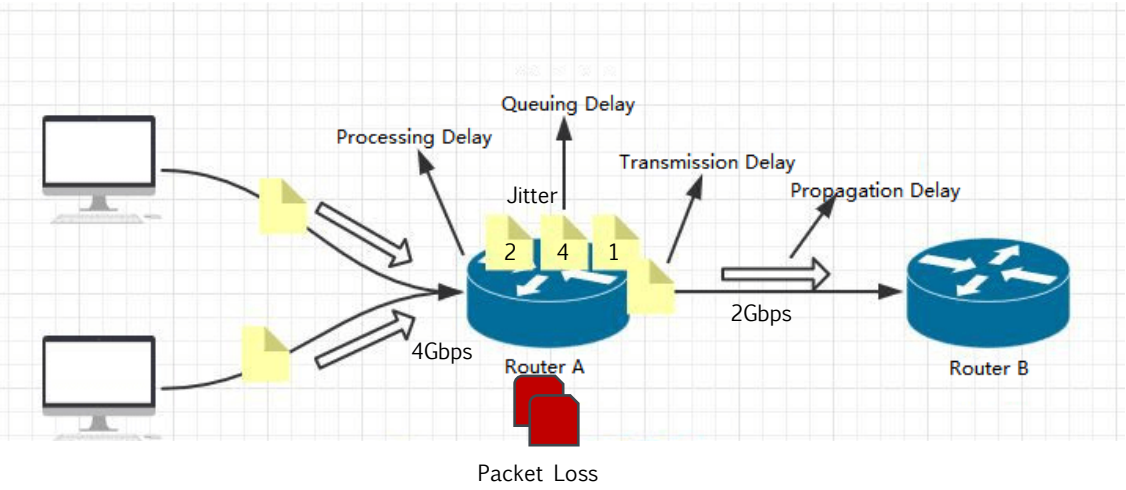
By the end of this chapter, students will be able to:

- Define QoS and explain why it's needed in modern networks.
- Identify and measure QoS performance metrics (delay, jitter, loss, bandwidth).
- Understand QoS models and mechanisms used in enterprise networks.
- Configure and verify QoS tools (classification, marking, queuing, policing).
- Evaluate QoS strategies in different environments.

Quality of Service (QoS)

Definition:
Quality of Service (QoS) refers to a suite of technologies and techniques in networking that enables the management and prioritization of data traffic. It is essential for ensuring that different types of data flows receive the appropriate levels of bandwidth, delay, jitter, and packet loss control. QoS allows networks to differentiate between various applications and user requirements, thereby providing a more predictable performance across diverse traffic types.

Why QoS Matters:
- Modern networks handle multiple traffic types (voice, video, data, IoT).
- Real-time applications (VoIP, video conferencing) need guaranteed delivery.
- Without QoS, all traffic is treated equally — causing delays and poor performance.

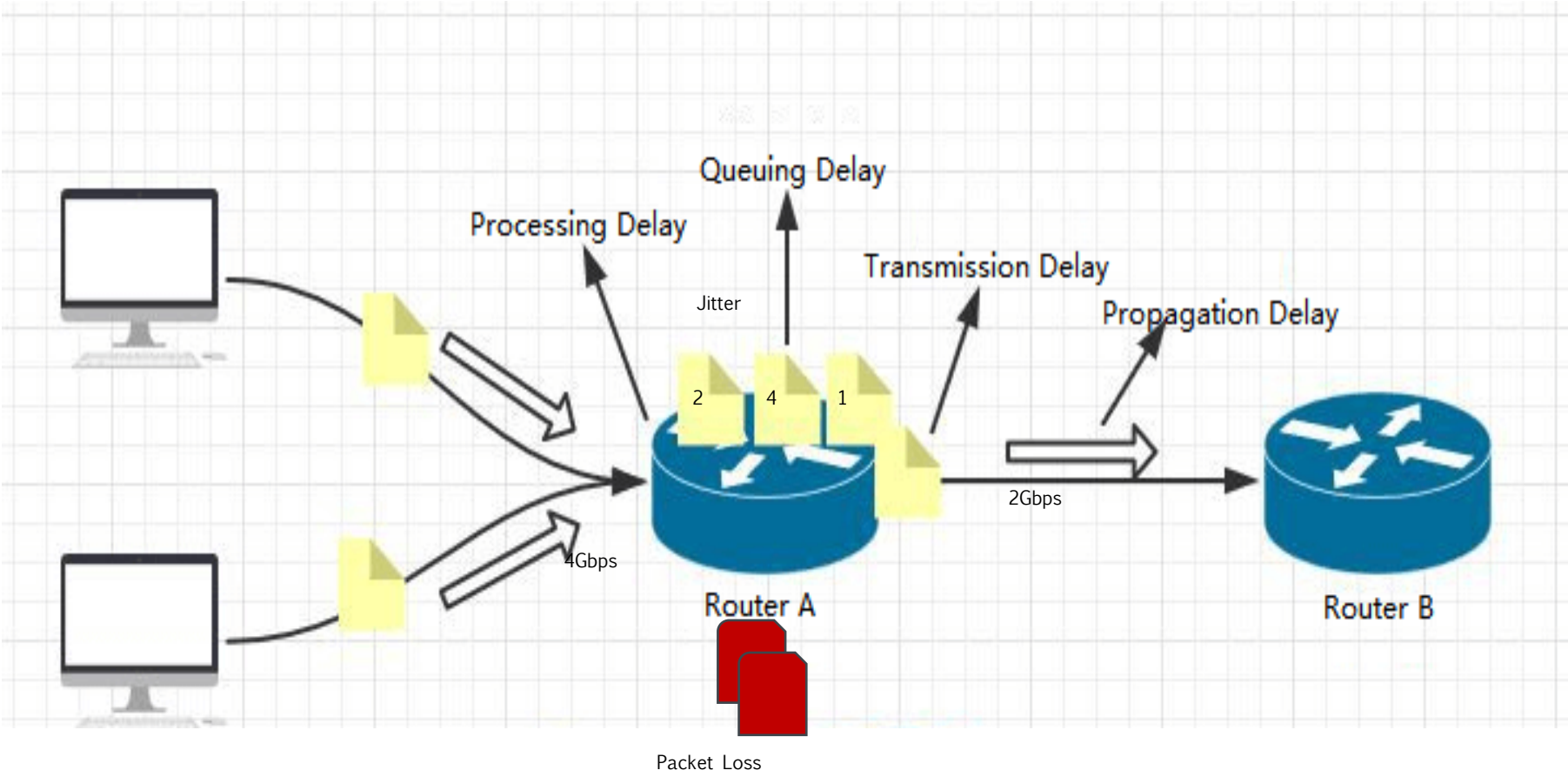


QOS PERFORMANCE METRICS

QoS relies on specific measurable network performance metrics:

Metric	Definition	Effect of Poor Value
Bandwidth	Capacity of a link (bps)	Congestion, slow transmission
Delay (Latency)	Time for a packet to reach destination	Echo in calls, video lag
Jitter	Variation in delay between packets	Choppy voice or video
Packet Loss	% of packets not delivered	Missing frames, distorted audio
Reliability	Network's ability to maintain connection	Service interruptions

Quality of Service (QoS)



QoS Tools and Mechanisms

Definition:
QoS tools are the set of mechanisms used by network devices (routers, switches) to control traffic behavior — by classifying, prioritizing, scheduling, delaying, or discarding packets — in order to meet specific performance requirements (latency, jitter, loss).

Main Goal:
Ensure that critical applications (e.g., voice, video conferencing, or control signals) receive better service than non-critical traffic (e.g., email, file downloads).

QoS Tools and Mechanisms

- Classification and Marking
- Queuing (Congestion Management)
- Congestion Avoidance
- Traffic Policing
- Traffic Shaping
- Link Efficiency Mechanisms (LEM)

MAIN QOS MECHANISMS

Classification and Marking

Queuing (Scheduling techniques)

Congestion Avoidance

Traffic Policing

Traffic Shaping

Link Efficiency Mechanisms (LEM)

Classification

Purpose: Identify and group packets based on specific characteristics.

How it works:

When a packet arrives at a router or switch, the device examines packet headers to determine the type of traffic and the application it belongs to.

- Source/destination IP address
- Protocol type (TCP/UDP)
- Port number (e.g., 80 for HTTP, 5060 for SIP)

Packets are placed into traffic classes (e.g., Voice, Video, Best Effort, Background).

Marking

Purpose: Tag packets with a priority value that other network devices can recognize and act upon.

Where it happens: Usually at the network edge — as close as possible to the traffic source (e.g., access switch or router).

How it works:

The device sets a value in a specific header field:

- Layer 2: CoS (Class of Service) — uses the 3-bit priority field in IEEE 802.1Q VLAN tag.
- Layer 3: DSCP (Differentiated Services Code Point) — uses 6 bits in the IP header.
- Layer 3: IP Precedence (older 3-bit field before DSCP).

MAIN QOS MECHANISMS

Classification and Marking

Queuing (Scheduling techniques)

Congestion Avoidance

Traffic Policing

Traffic Shaping

Link Efficiency Mechanisms (LEM)

Purpose:
When multiple packets compete for the same output interface, queuing mechanisms decide **which packet leaves next** — ensuring critical traffic is not delayed or dropped.
How it works:
Each router interface has buffers (queues).
When congestion occurs, packets are placed into these queues based on their priority class.
Example in practice:
Imagine a link carrying:

- Voice (priority queue)
- Video (medium class)
- Web traffic (best effort)

During congestion:

- Voice traffic always leaves first (low delay, low jitter).
- Video traffic gets the next priority.
- Web traffic waits or is dropped if necessary

MAIN QOS MECHANISMS

Classification and Marking

Queuing (Scheduling techniques)

Congestion Avoidance

Traffic Policing

Traffic Shaping

Link Efficiency Mechanisms (LEM)

Algorithm	Description	Use Case
FIFO (First In, First Out)	Simplest method — packets are processed in arrival order.	Not suitable for real-time applications.
PQ (Priority Queuing)	Packets are divided into priority queues (high, medium, low); higher ones always transmitted first.	VoIP, real-time apps.
WFQ (Weighted Fair Queuing)	Dynamically assigns bandwidth fairly based on weight (importance).	Balanced environments with mixed traffic.
CBWFQ (Class-Based WFQ)	Admin defines specific classes and assigns bandwidth shares explicitly.	Enterprise WAN links.
LLQ (Low-Latency Queuing)	Combines CBWFQ and PQ; offers strict priority for delay-sensitive traffic like voice.	VoIP and interactive video.

MAIN QOS MECHANISMS

Classification and Marking

Queuing (Scheduling techniques)

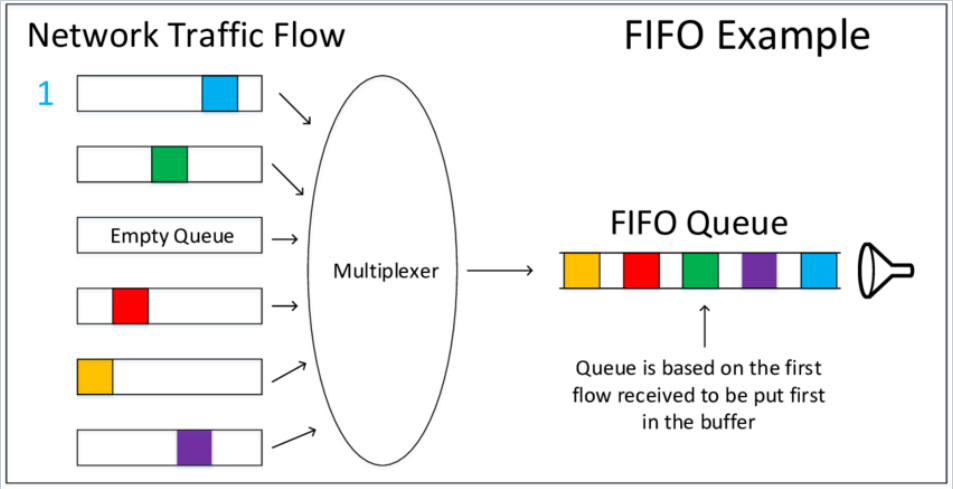
Congestion Avoidance

Traffic Policing

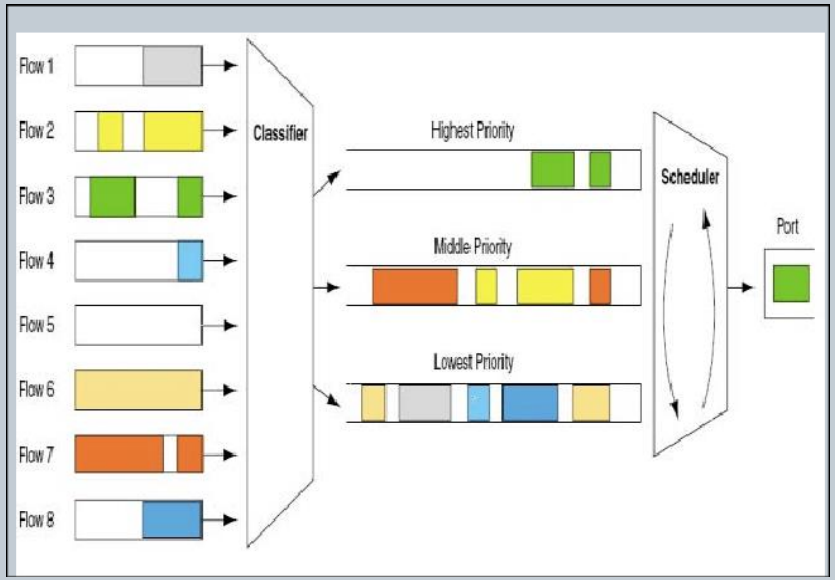
Traffic Shaping

Link Efficiency Mechanisms (LEM)

FIFO (First In, First Out)



PQ (Priority Queuing)



MAIN QOS MECHANISMS

Classification and Marking

Queuing (Scheduling techniques)

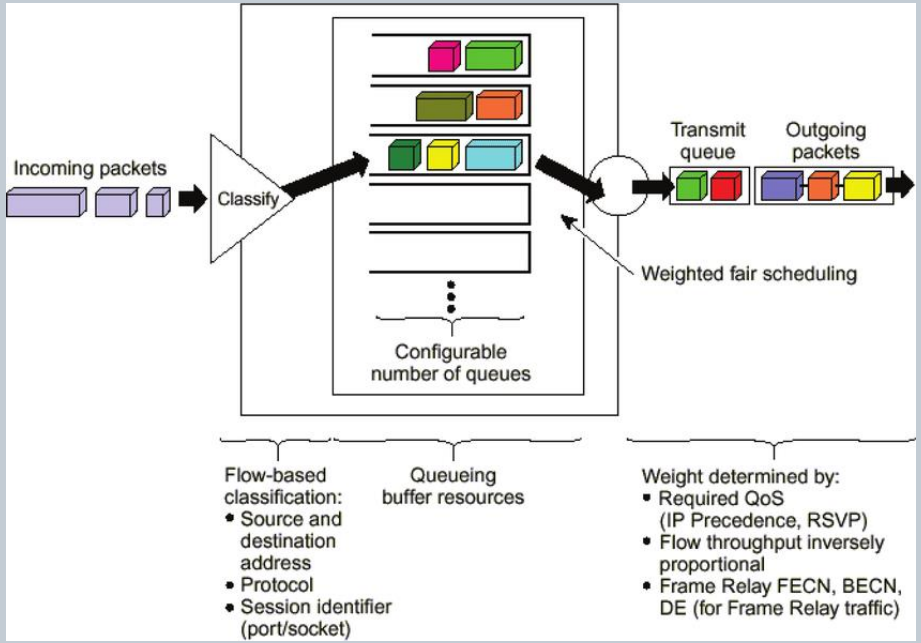
Congestion Avoidance

Traffic Policing

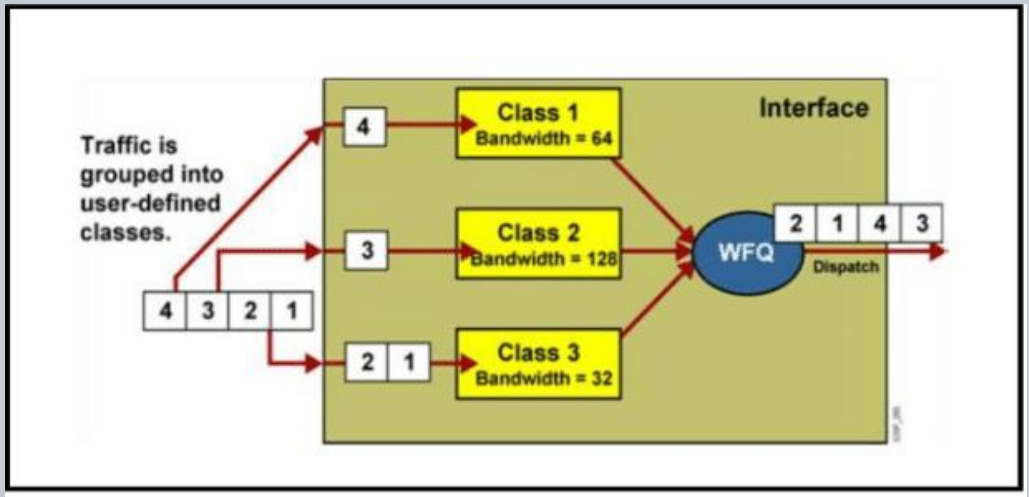
Traffic Shaping

Link Efficiency Mechanisms (LEM)

WFQ (Weighted Fair Queuing)



CBWFQ (Class-Based WFQ)



MAIN QOS MECHANISMS

Classification and Marking

Queuing (Scheduling techniques)

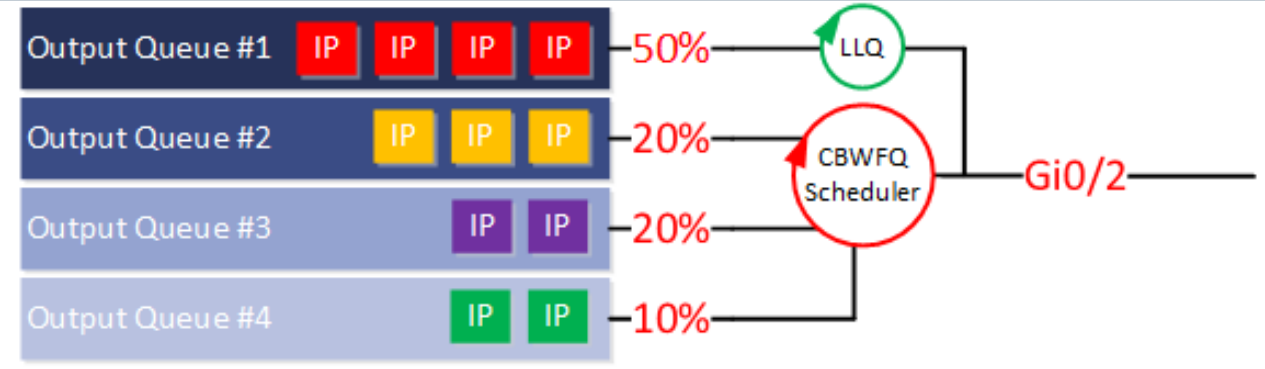
Congestion Avoidance

Traffic Policing

Traffic Shaping

Link Efficiency Mechanisms (LEM)

LLQ (Low-Latency Queuing)



MAIN QOS MECHANISMS

Classification and Marking

Queuing (Scheduling techniques)

Congestion Avoidance

Traffic Policing

Traffic Shaping

Link Efficiency Mechanisms (LEM)

Purpose:
Prevent network congestion *before* it happens by controlling buffer usage and signaling traffic sources to slow down.
How it works:
Routers monitor queue lengths; if queues start filling up, they begin selectively dropping packets rather than waiting for the queue to overflow completely.
Key Mechanisms:
RED (Random Early Detection):

- Calculates an *average queue size*.
- If queue exceeds a minimum threshold, packets are dropped randomly.
- This acts as an **early warning** to TCP sources to reduce their sending rate.

WRED (Weighted RED):
Enhanced version of RED.
Uses **different thresholds** for different classes of traffic (based on DSCP or IP precedence).
Example:

High-priority packets dropped later (higher threshold).
Low-priority packets dropped sooner.

MAIN QOS MECHANISMS

Classification and Marking

Queuing (Scheduling techniques)

Congestion Avoidance

Traffic Policing

Traffic Shaping

Link Efficiency Mechanisms (LEM)

Purpose:
Ensure that traffic does not exceed a predefined bandwidth limit.
Used for **rate enforcement** — not smoothing.
How it works:
Based on the **Token Bucket Algorithm**:
•Tokens accumulate in a bucket at a constant rate.
•Each packet consumes tokens based on its size.
•If enough tokens are available → packet transmitted.
•If not → packet is **dropped** or **remarked** to a lower priority.
Use cases:
•Enforcing Service Level Agreements (SLAs) on customer traffic.
•Preventing one flow from monopolizing a link.
Example:
•Voice VLAN limited to 1 Mbps.
If exceeded, extra packets dropped.

MAIN QOS MECHANISMS

Classification and Marking

Queuing (Scheduling techniques)

Congestion Avoidance

Traffic Policing

Traffic Shaping

Link Efficiency Mechanisms (LEM)

Purpose:
Regulate traffic flow by **buffering** packets and sending them at a steady rate — smoothing bursts rather than dropping packets.
How it works:
Also based on the **Token Bucket** model, but instead of dropping packets when tokens are unavailable, shaping **delays** them in a queue until tokens refill.

Aspect	Policing	Shaping
Excess packets	Dropped or remarked	Buffered and delayed
Goal	Enforce limit	Smooth flow
Best for	Core networks	Edge routers / WAN links

MAIN QOS MECHANISMS

Classification and Marking

Queuing (Scheduling techniques)

Congestion Avoidance

Traffic Policing

Traffic Shaping

Link Efficiency Mechanisms (LEM)

Purpose:
Improve efficiency and reduce delay on **low-bandwidth links**, such as satellite or remote branch connections.
Techniques:
1. Link Fragmentation and Interleaving (LFI)
•Large packets are fragmented into smaller ones.
•Small, time-sensitive packets (e.g., voice) are interleaved between fragments.
•This reduces delay (serialization delay) for voice packets.
2. Header Compression
•Removes redundant header information from IP packets (e.g., in VoIP flows).
•Example: cRTP (Compressed Real-Time Transport Protocol) reduces header from 40 bytes → ~2–4 bytes.
3. Payload Compression
•Compresses payload data to save bandwidth on low-speed links.

QoS Models

QoS models define how a network manages traffic and provides different service levels to various applications or users. In other words, it is the architectural frameworks or philosophies that define how and where QoS mechanisms are applied across a network.
There are three classical models for implementing QoS:

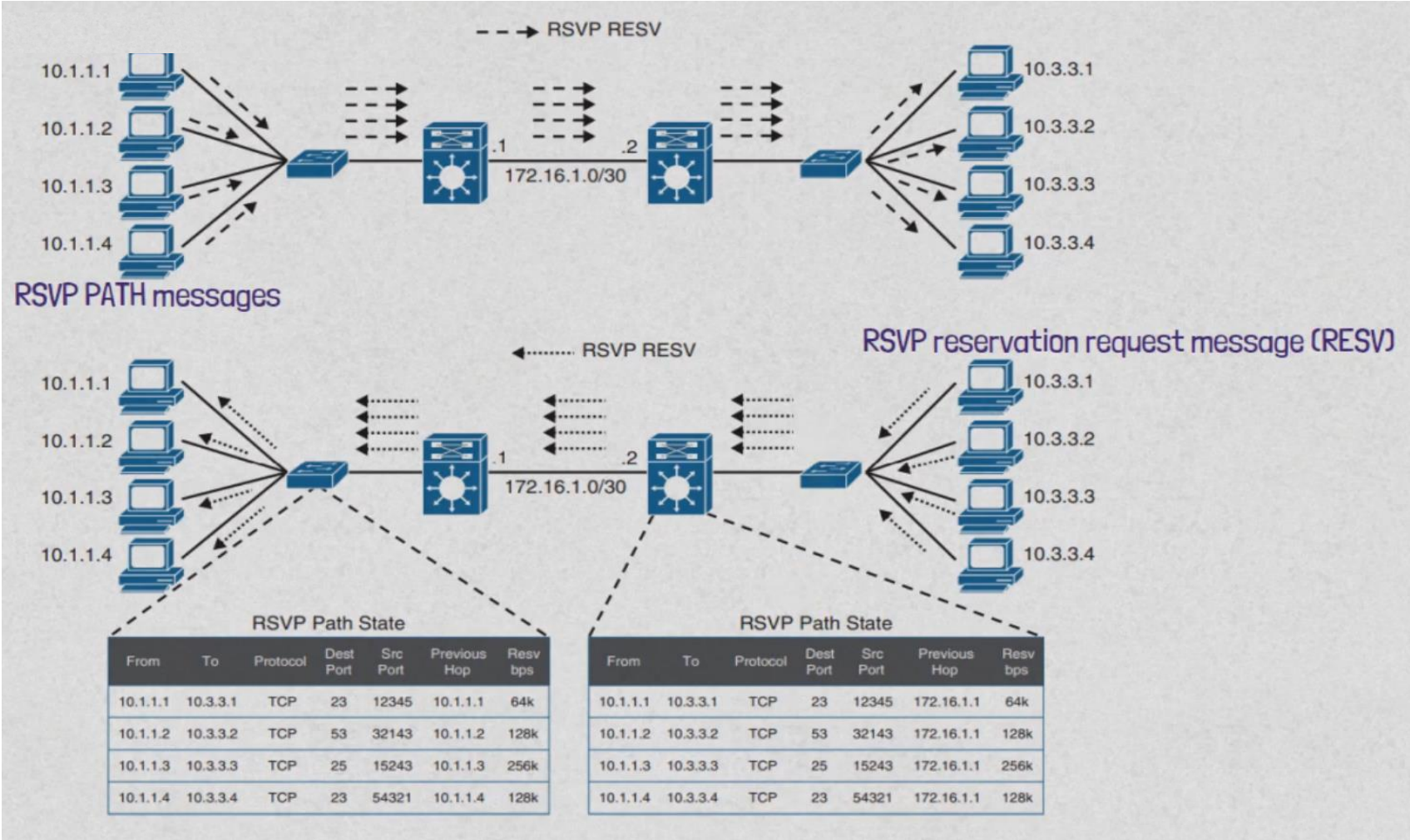
- Best Effort
- Integrated Services (IntServ)
- Differentiated Services (DiffServ)
- Multiprotocol Label Switching (MPLS)

QoS Key Concepts for marking and signaling mechanisms to identify, prioritize, and control traffic across the network.

- RSVP (Resource Reservation Protocol)
- CoS (Class of Service)
- ToS (Type of Service)
- DSCP (Differentiated Services Code Point)

Resource Reservation Protocol (RSVP)

A control protocol used to reserve network resources to provide quality of service (QoS) for real-time applications like video conferencing



From	To	Protocol	Dest Port	Src Port	Previous Hop	Resv bps
10.1.1.1	10.3.3.1	TCP	23	12345	10.1.1.1	64k
10.1.1.2	10.3.3.2	TCP	53	32143	10.1.1.2	128k
10.1.1.3	10.3.3.3	TCP	25	15243	10.1.1.3	256k
10.1.1.4	10.3.3.4	TCP	23	54321	10.1.1.4	128k

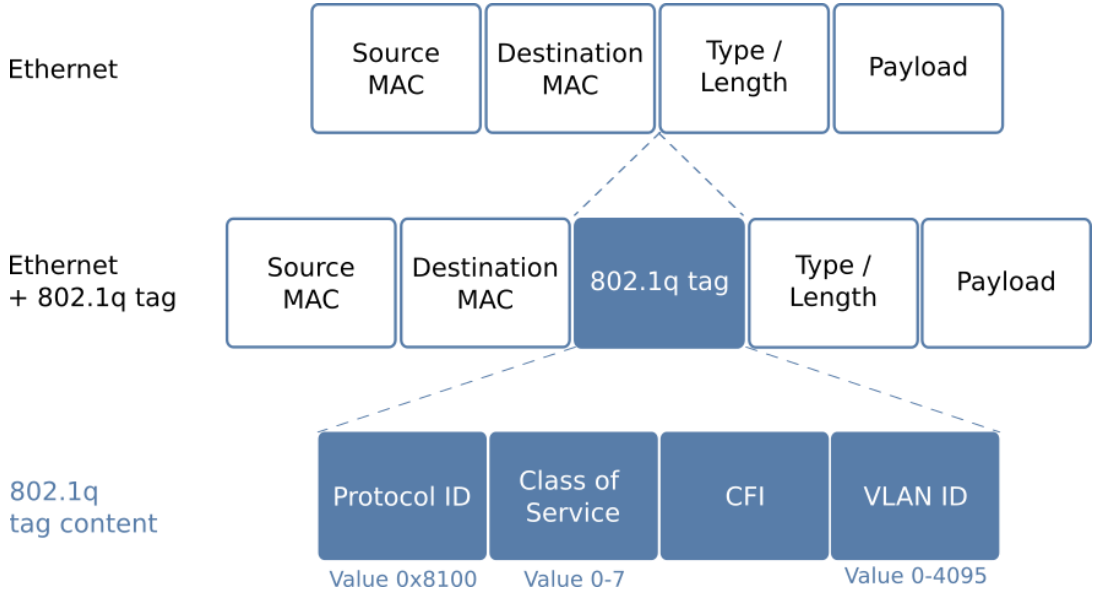
From	To	Protocol	Dest Port	Src Port	Previous Hop	Resv bps
10.1.1.1	10.3.3.1	TCP	23	12345	172.16.1.1	64k
10.1.1.2	10.3.3.2	TCP	53	32143	172.16.1.1	128k
10.1.1.3	10.3.3.3	TCP	25	15243	172.16.1.1	256k
10.1.1.4	10.3.3.4	TCP	23	54321	172.16.1.1	128k

Class of Service (CoS)

- Defined in Layer 2 (Data Link Layer) – IEEE 802.1Q VLAN tag.
- Uses 3 bits called Priority Code Point (PCP) in the VLAN header → allows 8 priority levels (0–7).
- Classifies traffic before it reaches Layer 3 (IP).

Typical values:

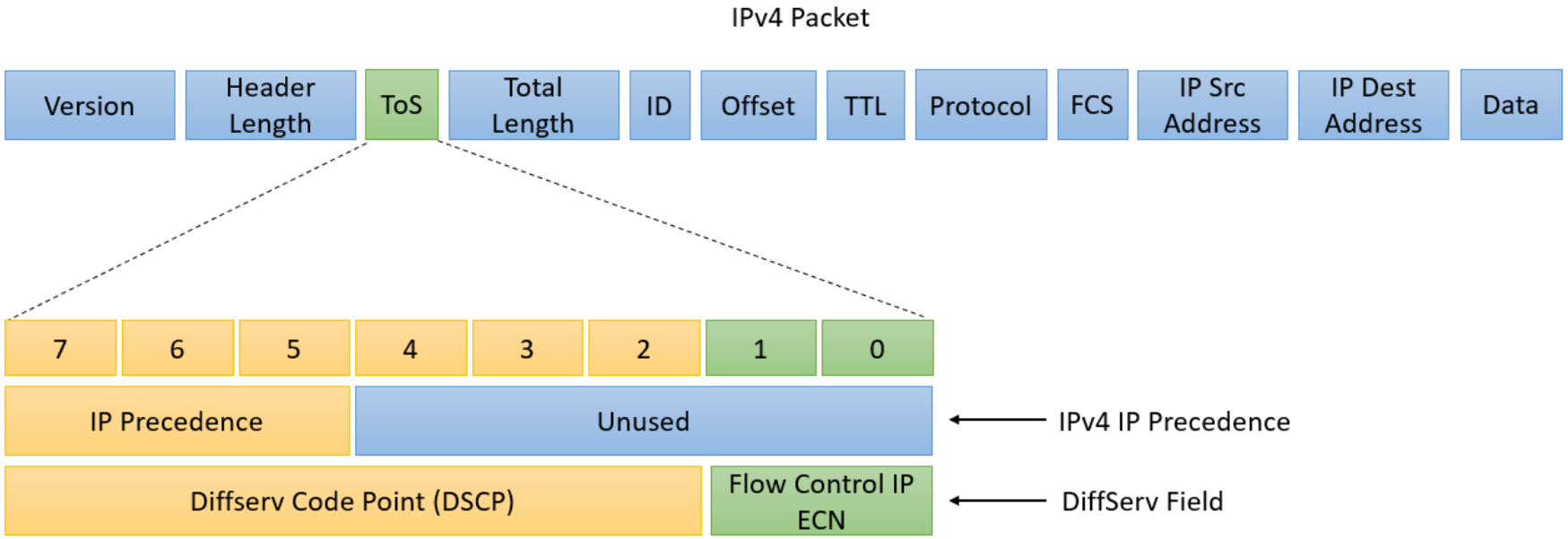
- 6-7 reserved for Network control
- 5 reserved for Voice over IP (high priority)
- 0 reserved for Best effort



Type of Service (ToS)

Defined in the **IPv4 header (8-bit field)**.
Originally used for prioritizing IP packets by delay, throughput, and reliability.
Structure:

- **3 bits precedence** → IPP
- **3 bits ToS flags** → DSCP
- **2 bit unused** → ECN



Differentiated Services Code Point (DSCP)

PHB (Per-Hop Behavior)	Decimal Value	Binary (6 bits)
Default (Best Effort)	0	000000
CS1 (Class Selector 1)	8	001000
CS2 (Class Selector 2)	16	010000
CS3 (Class Selector 3)	24	011000
CS4 (Class Selector 4)	32	100000
CS5 (Class Selector 5)	40	101000
CS6 (Class Selector 6)	48	110000
CS7 (Class Selector 7)	56	111000
EF (Expedited Forwarding)	46	101110

DSCP Assured Forwarding Values

PHB (Per-Hop Behavior)	Description (priority/drop)	Decimal Value	Binary
AF11 (Assured Forwarding 1 Low)	Low priority-Low drop probability	10	001010
AF12 (Assured Forwarding 1 Medium)	Low priority-Medium drop precedence	12	001100
AF13 (Assured Forwarding 1 High)	Low priority-High drop precedence	14	001110
AF21 (Assured Forwarding 2 Low)	Medium priority-Low drop probability	18	010010
AF22 (Assured Forwarding 2 Medium)	Medium priority-Medium drop precedence	20	010100
AF23 (Assured Forwarding 2 High)	Medium priority-High drop precedence	22	010110
AF31 (Assured Forwarding 3 Low)	High priority-Higher reliability traffic	26	011010
AF32 (Assured Forwarding 3 Medium)	High priority-Medium drop precedence	28	011100
AF33 (Assured Forwarding 3 High)	High priority-High drop precedence	30	011110
AF41 (Assured Forwarding 4 Low)	Premium applications (e.g., video)	34	100010
AF42 (Assured Forwarding 4 Medium)	Very high priority-Medium drop precedence	36	100100
AF43 (Assured Forwarding 4 High)	Very high priority-High drop precedence	38	100110

Best Effort Model (No QoS)

- It is the default QoS model used for Internet and it doesn't implement any QoS mechanism at all, that is the reason why there isn't any complexity associated to this QoS model.
- "send and hope for the best."
- All packets are treated equally; no prioritization or resource reservation.

How It Works

- The network forwards packets based on available resources.
- If congestion happens, packets are dropped randomly.

Characteristics

- Easy to implement.
- Low overhead (no QoS setup needed).
- No certainty about delay, loss, or throughput.
- Performs poorly for real-time traffic (voice, video).

Integrated Services (IntServ)

It's a model based on flows, i.e., source and destination IP addresses and ports.

applications ask to the network for an explicit resource reservation per flow. Network devices keep track of all the flows traversing the nodes checking if new packets belong to an existing flow and if there are enough network resources available to accept the packet.

The implementation of this model requires the presence of IntServ capable routers in the network and uses RSVP for end-to-end resource reservation. RSVP enables a host to establish a connection over connectionless IP Internet:

- Applications request some level of service to the network before sending data.
- The network admits or rejects the reservation (per flow) based on available resources.
- Once cleared, the network expects the application to remain within the requested traffic profile.

The scalability of this model is limited by the fact that exists a high resource consumption on network nodes caused by per flow processing and associated state. Remember that network nodes need to maintain the reservation state for each flow traversing the node.

Differentiated Services (DiffServ)

Differentiated Services (Diffserv) model is also known as a soft QoS model. It's a model based in service classes and per hop behaviours associated to each class.

In this case, there is no need for an explicit request for resource reservation by applications to the network. Differentiated Services is based on statistical preferences per traffic class.

DiffServ allows end devices or hosts to classify packets into different treatment categories or Traffic Classes (TC), each of which will receive a different Per-Hop-Behaviour (PHB) at each hop from the source to the destination. Each network device on the path treats packets according to the locally defined PHB.

Priorities are marked in each packet using DSCP for traffic classification. This marking is performed per packet usually at the QoS domain boundary. The marking can be done at several levels of the networking layers (MPLS EXP, CoS).

Chapter 04

Emerging Network Technologies

Objectives: by the end of this chapter, students will be able to:

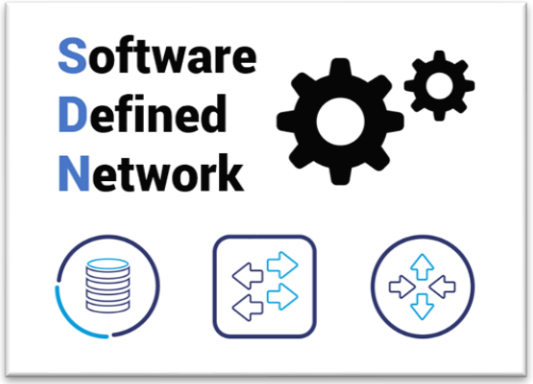
- Understand SDN, NFV, and IoT architectures.
- Identify key protocols and control mechanisms.
- Explain how these technologies integrate in modern networks.

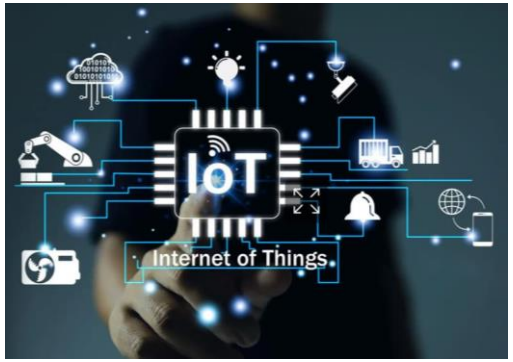
What Are Emerging Network Technologies?

Emerging Network Technologies refer to new and evolving approaches in networking that aim to overcome the limitations of traditional networks — such as rigidity, manual configuration, scalability issues, and lack of automation.

They leverage virtualization, programmability, and intelligence to make networks more flexible, efficient, and adaptable to modern demands like 5G, IoT, and cloud computing.

Traditional networks were designed for fixed infrastructure, limited devices, and predictable traffic. However, today's networks must handle massive connectivity dynamic services high performance and reliability, To meet these challenges, new paradigms such as:

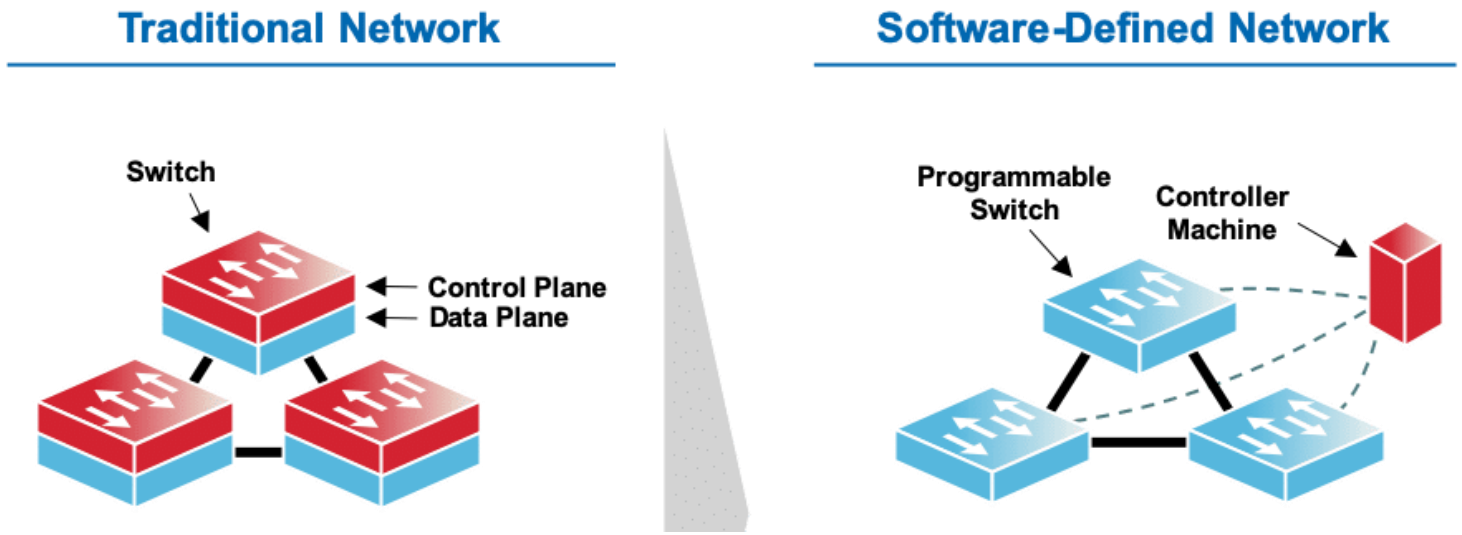




have emerged to reshape how networks are built, managed, and optimized.

Control plane and data plane

- The **control plane** is the "brain" of the network, it makes **decisions** about how packets should be handled. In SDN, this plane is **centralized** in a **controller** rather than distributed across devices.
- The **data plane** (or **forwarding plane**) is responsible for **moving packets** from one interface to another according to pre-defined rules.



Why SDN Separation Matters

- Simplifies management: Central control = easier configuration and troubleshooting.
- Programmability: Controllers expose APIs to automate network policies.
- Flexibility: Network behavior can change dynamically (e.g., prioritizing video traffic).
- Security: Central view allows for rapid detection and mitigation of attacks.

Software-Defined Networking (SDN)

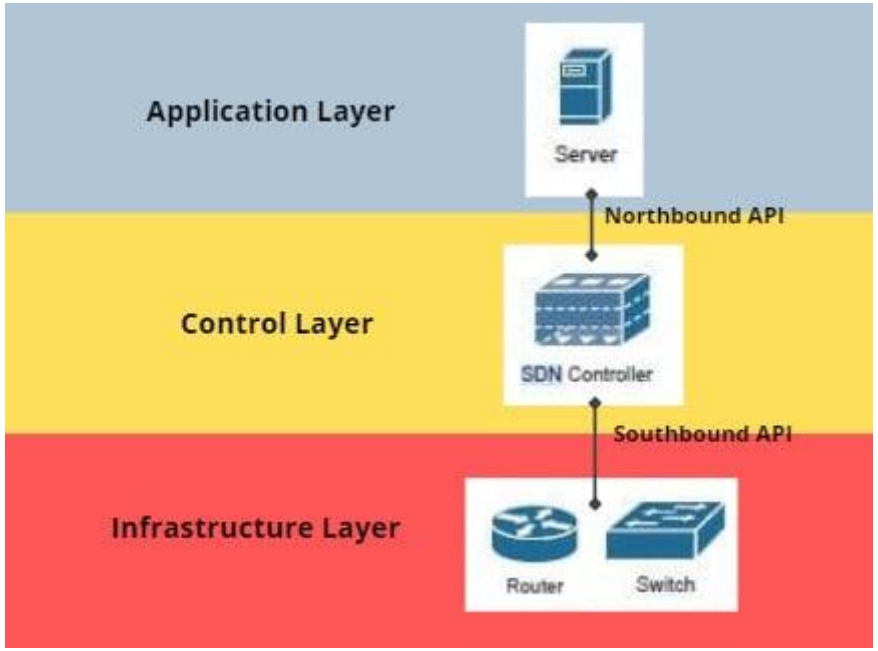
SDN separates the control plane (decision-making) from the data plane (packet forwarding). This separation enables centralized control, automation, and programmable management of the network.

Traditional vs. SDN Network:

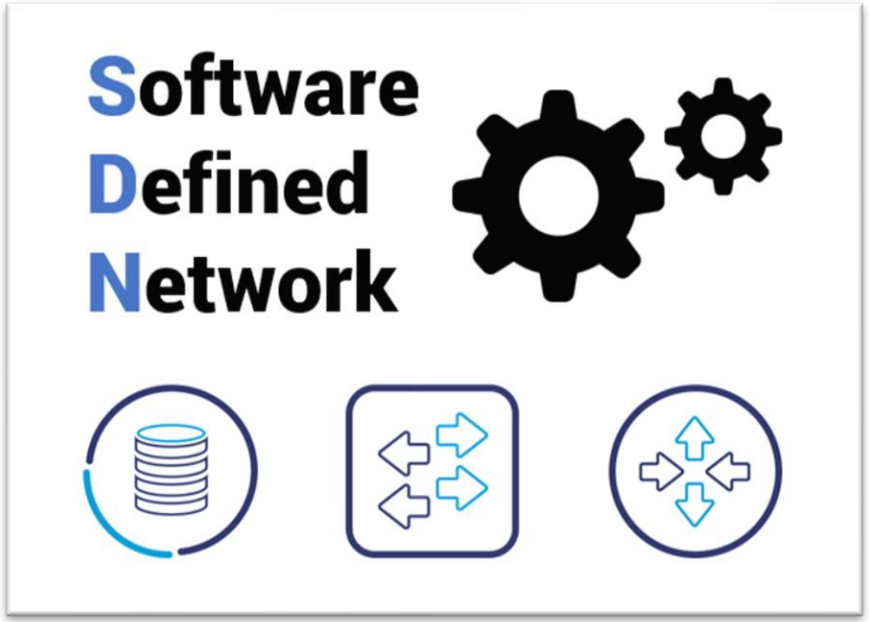
- Traditional: Each device (router/switch) makes its own decisions.
- SDN: A central controller dictates forwarding rules for all devices.

Key Components:

- Application Layer – Network applications (firewall apps, load balancers).
- Control Layer – SDN controller (e.g., OpenDaylight, ONOS).
- Infrastructure Layer – Physical/virtual switches and routers.



WHY SDN?



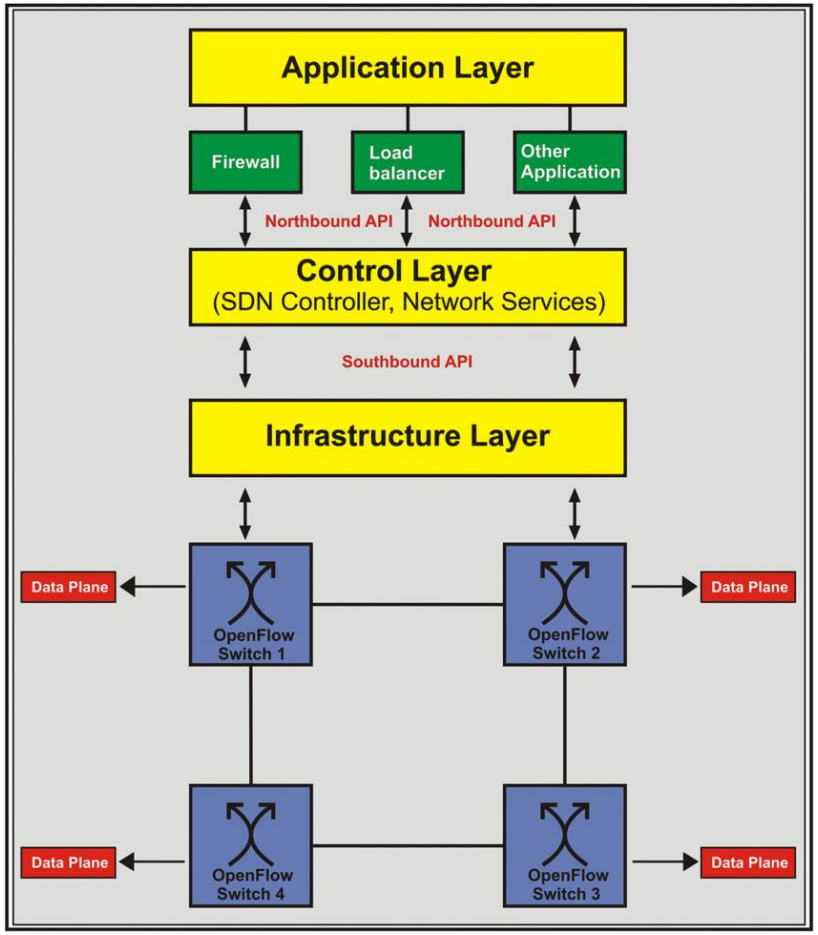
Limitations of Traditional Networking:

- Complex and error-prone configuration.
- Vendor lock-in (proprietary hardware and software).
- Slow adaptation to new policies or security requirements.

SDN Advantages:

- **Centralized management** via software controllers.
- **Programmability:** networks can be configured through APIs.
- **Agility:** quick deployment of new services.
- **Automation:** reduces manual errors.

SDN ARCHITECTURE (3-LAYER MODEL)



1.Application Layer

1. Contains network applications (e.g., load balancing, firewalls, routing).
2. Communicates with the controller via **Northbound APIs**.

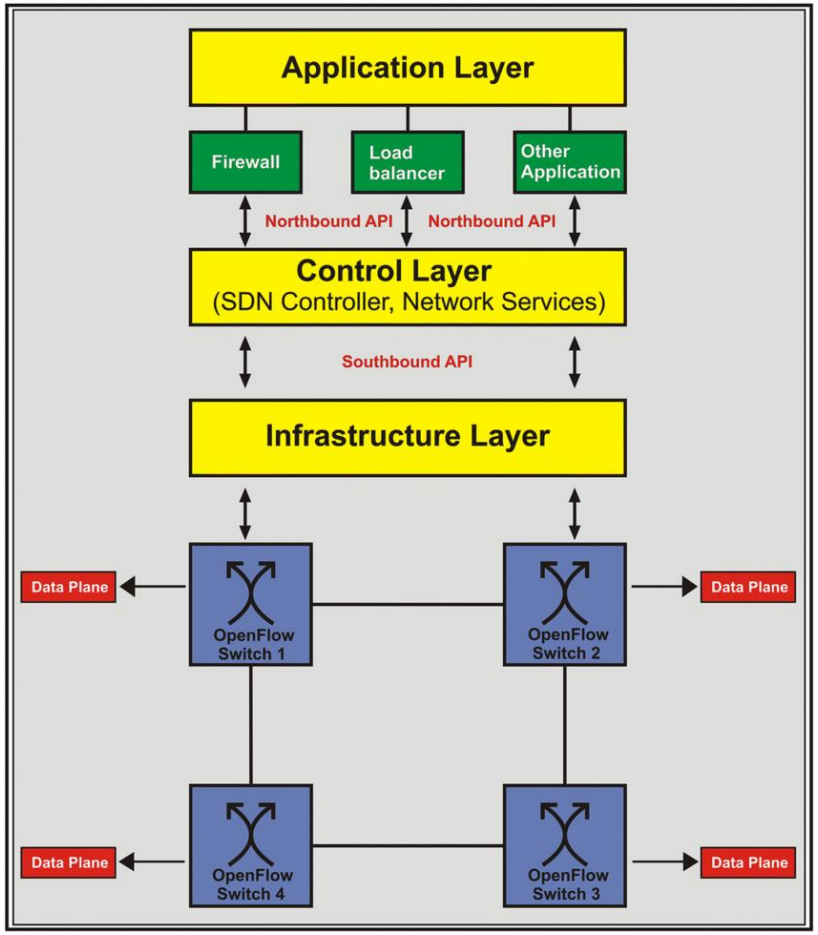
2.Control Layer

1. The **SDN Controller** acts as the network's "brain."
2. Uses **Southbound APIs** (like OpenFlow) to communicate with devices.

3.Infrastructure Layer

1. Physical and virtual switches and routers (the **data plane**).
2. Responsible for packet forwarding based on flow rules.

SDN INFRASTRUCTURE LAYER



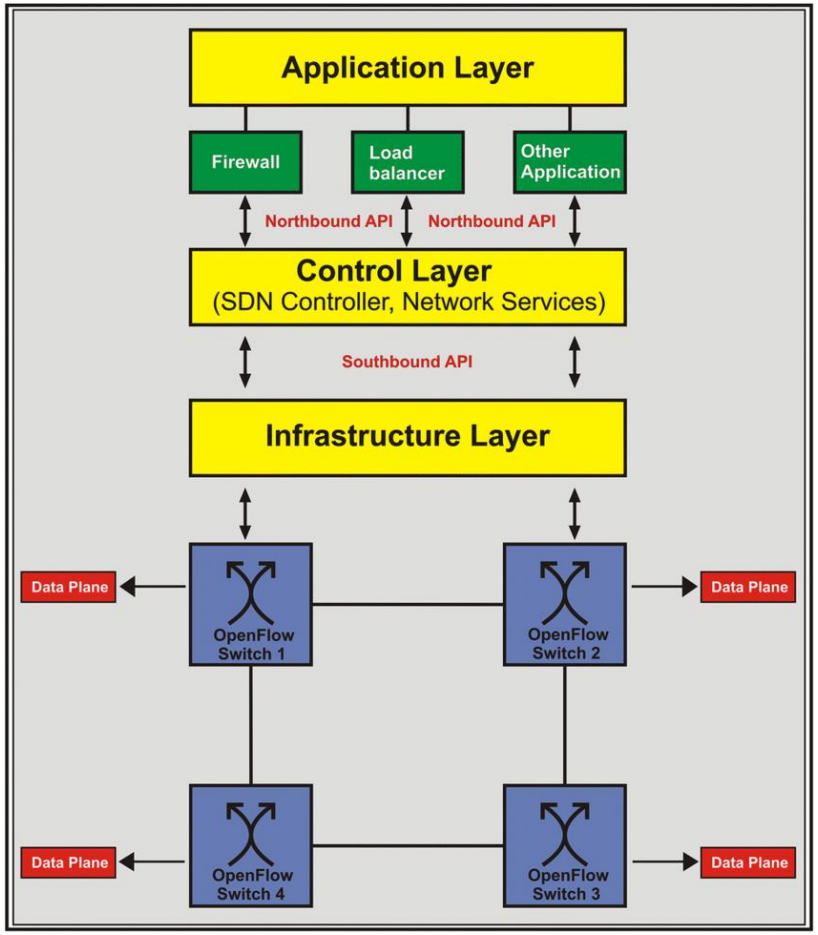
Role: Execute the instructions from the controller (forward, drop, modify, send).

Components:

- **OpenFlow Switches:** Hardware or software-based devices that understand controller instructions.
- **Flow Tables:** Contain match–action rules (if packet matches criteria → perform action).
- **Virtual Switches (vSwitch):** Run in hypervisors or cloud environments (e.g., Open vSwitch).

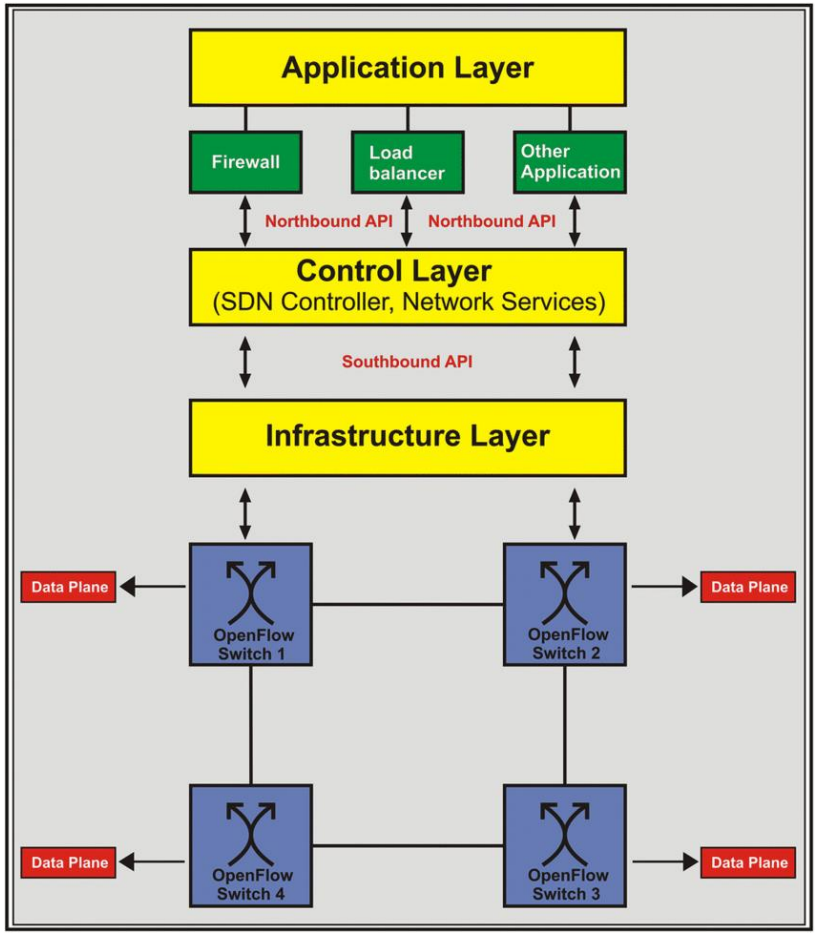
Protocol Example: OpenFlow, NETCONF, gNMI.

SDN CONTROL PLANE (THE CONTROLLER)



Function:
Central point that determines how traffic flows in the network.
Tasks:
•Maintains network topology.
•Installs, updates, or removes flow rules.
•Collects statistics from switches.
•Enforces policies (QoS, access control, routing).
Examples:
- **ONOS (Open Network Operating System)**
- **OpenDaylight**
- **Floodlight, Ryu, Cisco APIC**

SDN APPLICATION LAYER



Purpose:
Provides user-facing functionality that defines network behavior.

Common SDN Applications:

- **Traffic Engineering:** Optimize bandwidth usage.
- **Security Monitoring:** Intrusion detection, anomaly detection.
- **Load Balancing:** Distribute flows across multiple paths.
- **QoS Management:** Prioritize delay-sensitive traffic.

Interface:
Communicates with the controller via **Northbound APIs** (usually RESTful APIs).

OpenFlow Protocol (Southbound Interface)

OpenFlow is the **first and most widely used** SDN protocol that defines communication between the controller and forwarding devices.

How it Works:

1.Controller installs **flow entries** into switches.

2.Each entry includes:

1. Match fields (IP, MAC, VLAN, TCP port).
2. Actions (forward, drop, modify, send to controller).
3. Counters (statistics).

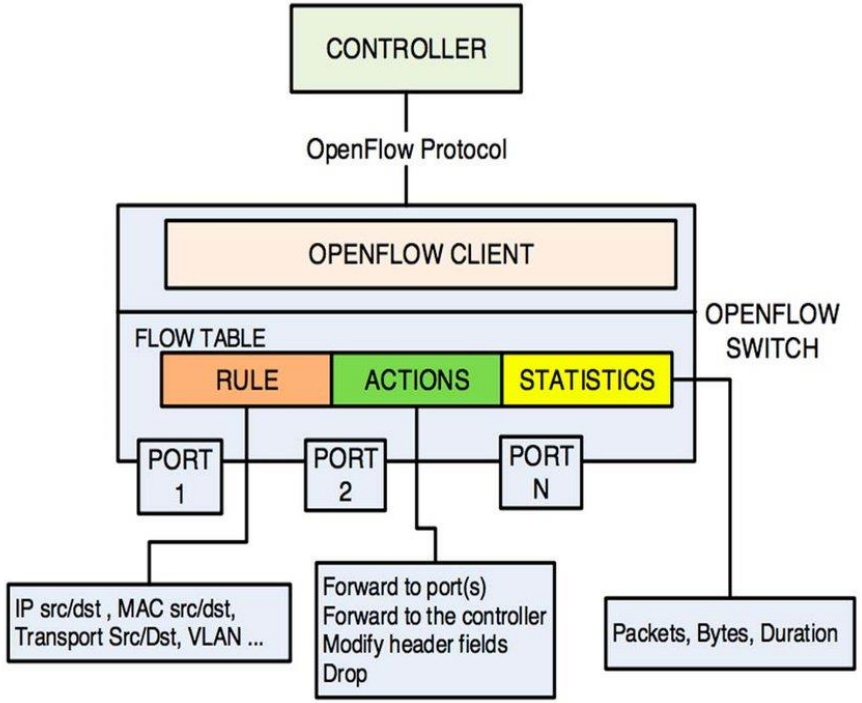
OpenFlow Messages:

•**Packet-In:** Switch → Controller (no matching rule found).

•**Flow-Mod:** Controller → Switch (install or delete a flow rule).

•**Packet-Out:** Controller → Switch (send specific packet).

•**Stats-Request/Reply:** For monitoring traffic flows.



Flow Table Operation

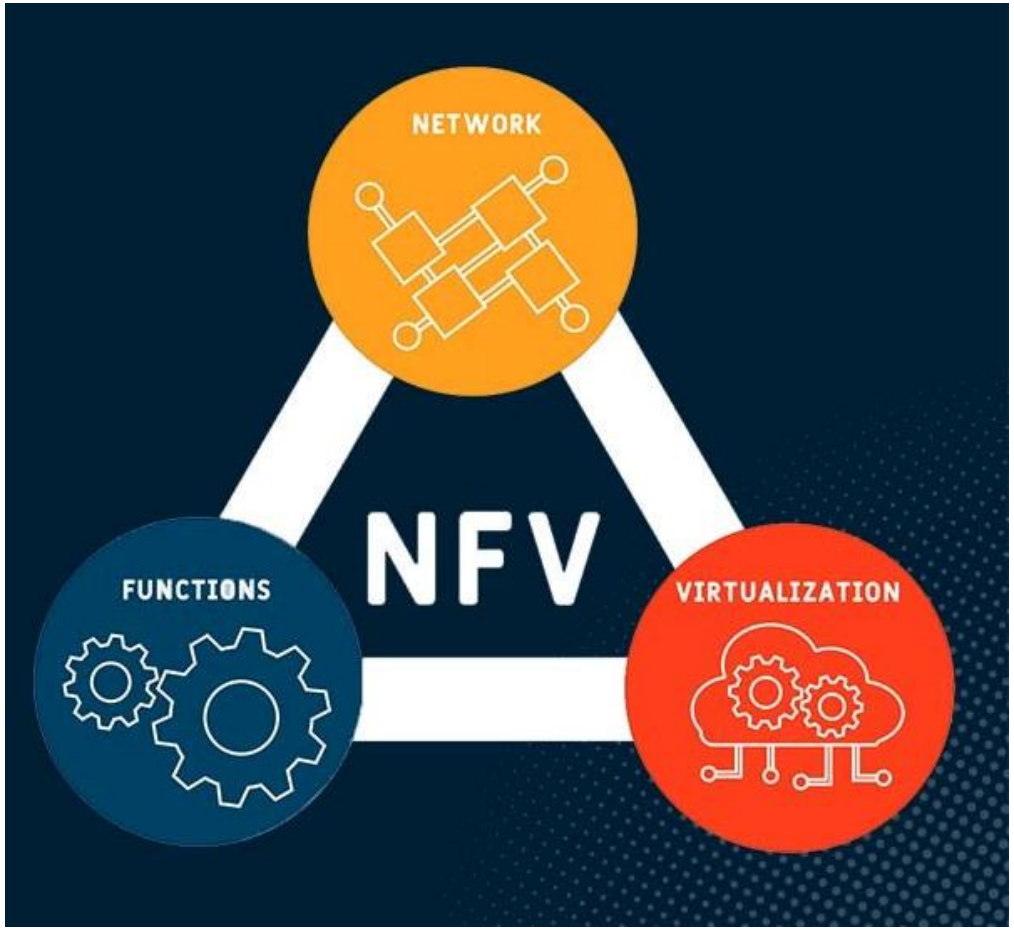
Flow Entry Example:

Match Field	Action	Priority
IP src = 192.168.1.10	Forward to port 3	10
TCP dst = 80	Forward to port 2	5
Any	Send to controller	1

The process:
1.Packet arrives → matched against flow entries.
2.Matching action executed.
3.If no match → packet sent to controller for decision.

Introduction to NFV

- Network Function Virtualization (NFV) is a technology, **Developed by the** European Telecommunications Standards Institute (ETSI) in 2012, that **decouples network functions** (e.g., firewall, NAT, DPI, load balancer) from proprietary hardware and runs them as **software on virtualized servers**
- Replace specialized network appliances with **Virtual Network Functions (VNFs)** running on **standard hardware**.



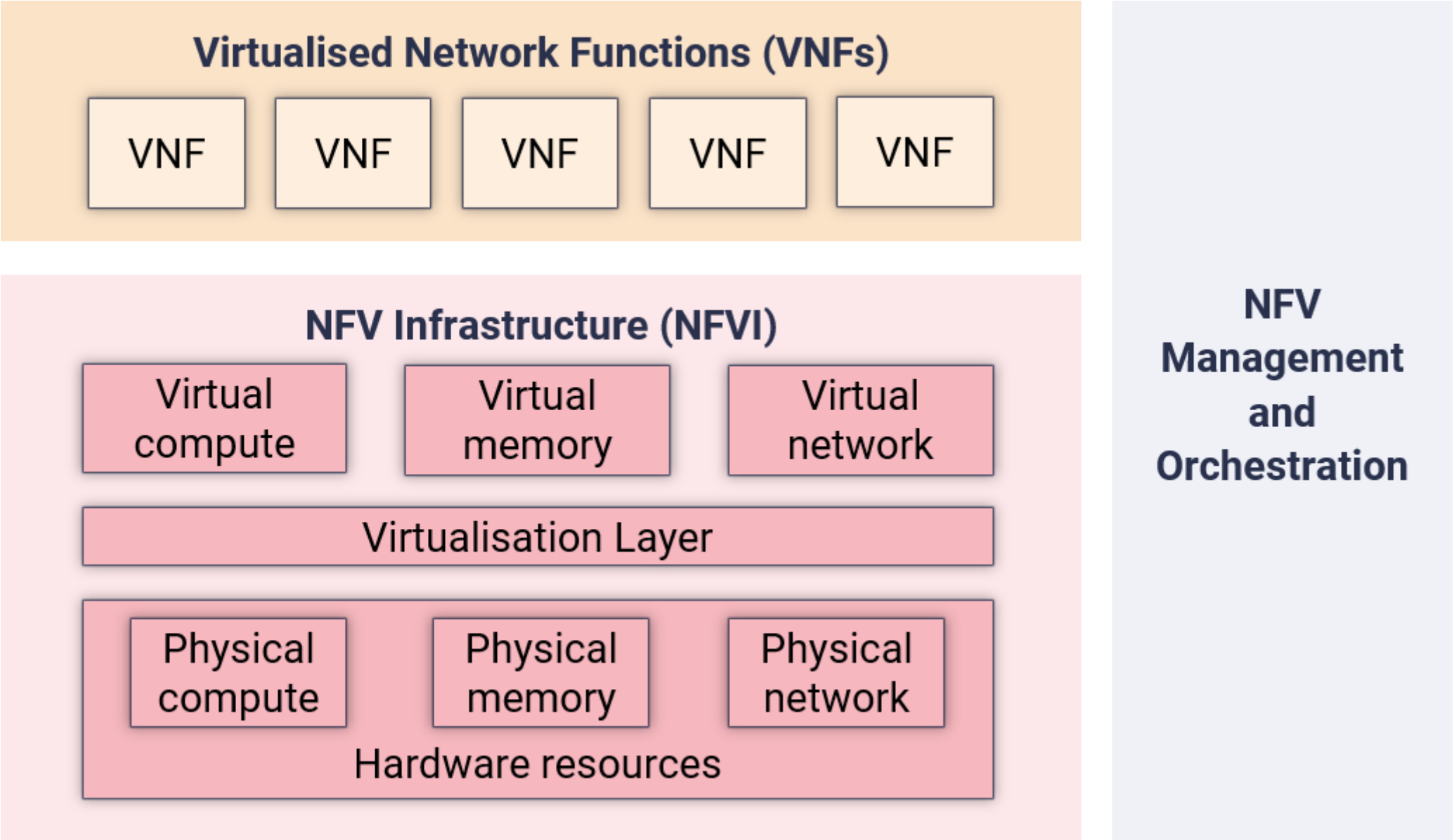
Motivation for NFV

Challenge	Explanation	Impact
High Capital and Operational Costs (CAPEX/OPEX)	Specialized hardware (e.g., Cisco ASR, Juniper MX, Palo Alto firewalls) is expensive to buy, power, and maintain.	Limits innovation and increases total cost of ownership.
Vendor Lock-In	Each appliance comes from a specific vendor with proprietary software and interfaces.	Difficult to integrate multi-vendor solutions or migrate.
Slow Service Deployment	Adding a new network service requires physically installing and configuring hardware.	Delays rollout of new services by weeks or months.
Limited Scalability and Flexibility	Scaling requires buying and installing new devices.	Poor elasticity during peak demand (e.g., traffic surges).
Inefficient Resource Utilization	Hardware often underutilized (only 30–40% capacity used).	Wasted energy and infrastructure.
Difficult Automation and Management	Manual configuration of each appliance.	High risk of errors and inconsistent policies.
Physical Space and Energy Constraints	Many appliances consume significant power and rack space.	Not suitable for cloud data centers or edge environments.

Motivation for NFV

NFV Concept	How it Solves the Problem	Example
Virtualized Network Functions (VNFs)	Run network functions (e.g., firewall, load balancer) as software on commodity servers.	Deploy a vFirewall on a virtual machine instead of a physical firewall.
Standard Hardware (x86/ARM)	Eliminates need for specialized equipment.	One server can host multiple VNFs.
Dynamic Scaling	VNFs can be started, stopped, or scaled automatically.	Increase bandwidth by deploying extra vRouters during peak time.
Centralized Management (MANO)	Automated orchestration replaces manual setup.	NFV Orchestrator deploys VNFs across data centers automatically.
Integration with SDN	Enables programmable connectivity between VNFs.	SDN controller reroutes traffic through VNFs dynamically.

NFV Architecture

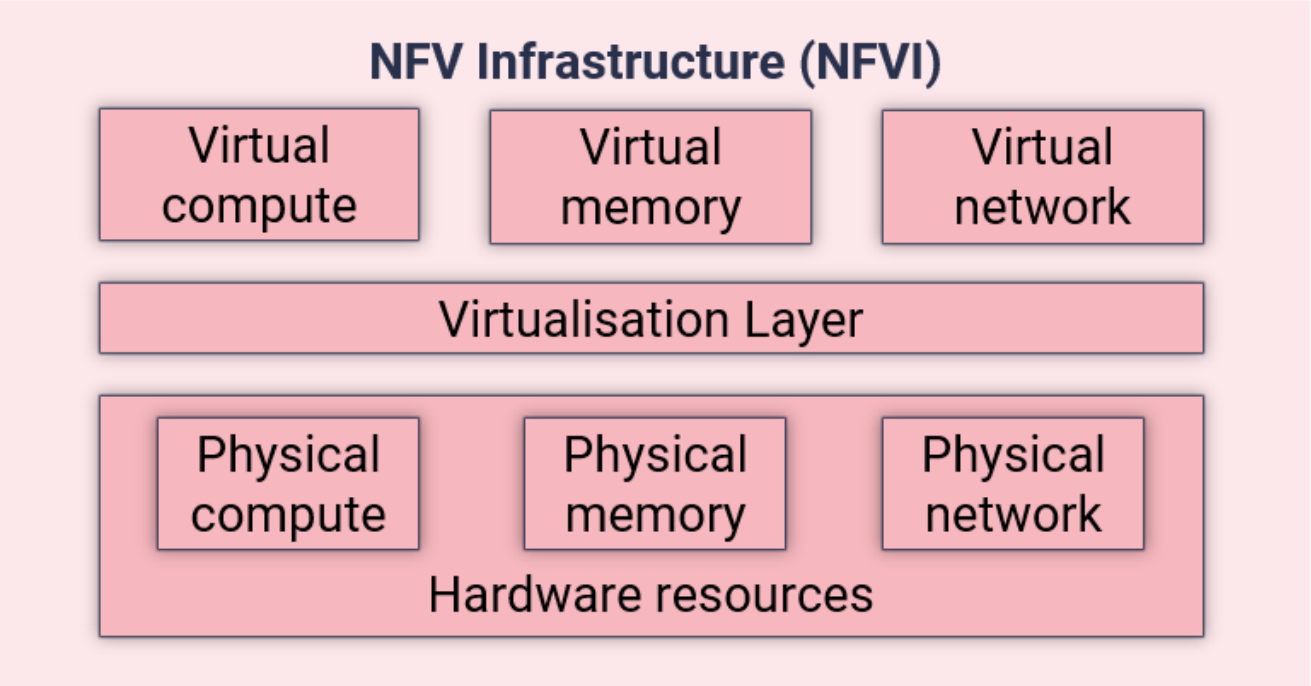


NFV Infrastructure (NFVI)

NFVI = Physical Resources + Virtualization Layer + Virtual Resources

- **Physical layer:** Standard servers, switches, storage systems.
- **Virtualization layer:** Uses **hypervisors** (KVM, VMware, Xen) or **containers** (Docker, Kubernetes).
- **Virtual resources:** Virtual CPUs, memory, virtual networks (vSwitches).

Goal: Provide the environment for VNFs to run dynamically and efficiently.

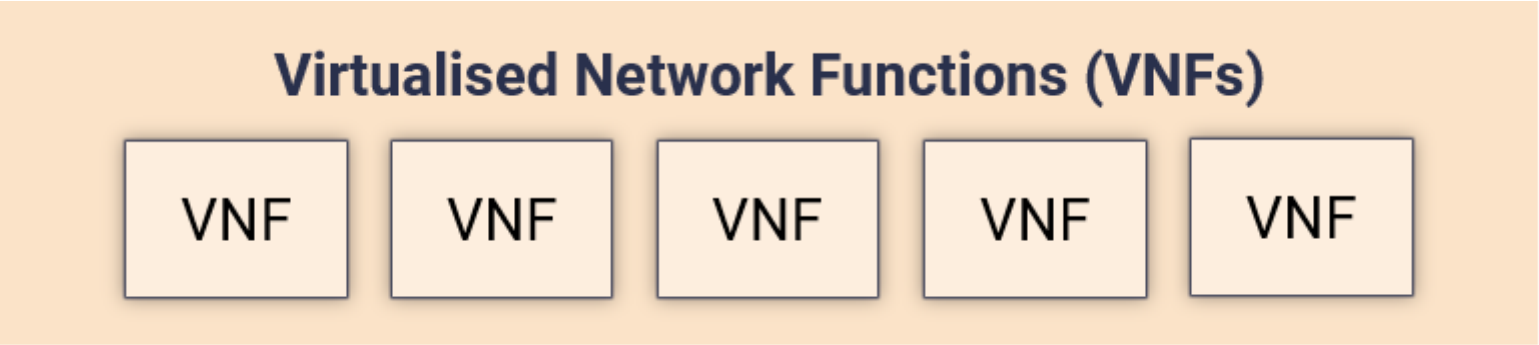


Virtual Network Functions (VNFs)

Each VNF provides a specific network service.
Examples:

- **vFirewall** → Security filtering
- **vRouter / vSwitch** → Packet forwarding
- **vDPI (Deep Packet Inspection)** → QoS / monitoring
- **vEPC (Evolved Packet Core)** → Mobile core in 4G/5G networks

VNFs can be **chained together** to build complete network services, called **Service Function Chaining (SFC)**.



Management and Orchestration (MANO)

NFV Orchestrator (NFVO)

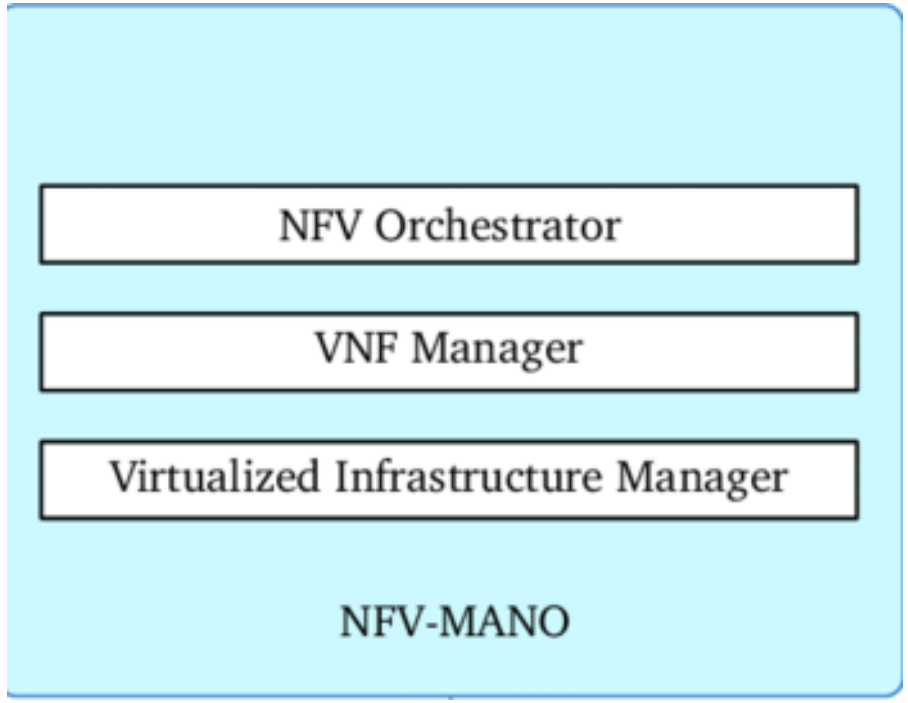
- Manages network services lifecycle and resource coordination.
- Interacts with SDN controllers for network setup.

VNF Manager (VNFM)

- Handles individual VNF instances (instantiation, scaling, updates).

Virtualized Infrastructure Manager (VIM)

- Manages compute, network, and storage resources (e.g., OpenStack).



NFV and SDN Integration

Software-Defined Networking (SDN) and **Network Function Virtualization (NFV)** are **complementary technologies** that, when combined, enable **flexible, programmable, and automated networks**.

- **NFV** virtualizes network services (e.g., firewall, router, load balancer).
- **SDN** provides centralized control and programmability of the network that connects those virtual functions.

Together, they form the foundation for **modern 5G, cloud, and edge infrastructures**.

"NFV **creates** the virtualized functions, and SDN **connects and manages** them intelligently."

Integration Architecture

A typical **SDN–NFV integrated architecture** includes three main layers:

- **Infrastructure Layer**

Consists of **NFV Infrastructure (NFVI)**: compute, storage, and network resources.

SDN switches (physical or virtual) forward packets based on instructions from the SDN controller.

Connectivity between VNFs is established dynamically using **SDN flows** (e.g., via OpenFlow or NETCONF).

- **Control Layer**

The **SDN Controller** provides centralized network control and exposes **northbound APIs** to higher layers.

The **NFV MANO** (Orchestrator, VIM, VNFM) manages VNF lifecycle and coordinates with the SDN controller for connectivity.

Communication between MANO and SDN uses **REST APIs** or **NETCONF/YANG**.

- **Application / Service Layer**

Network services are modeled as **Service Function Chains (SFC)** — ordered sequences of VNFs (e.g., firewall → IDS → load balancer).

The orchestrator deploys these chains while SDN ensures the traffic flows through them in the right order.

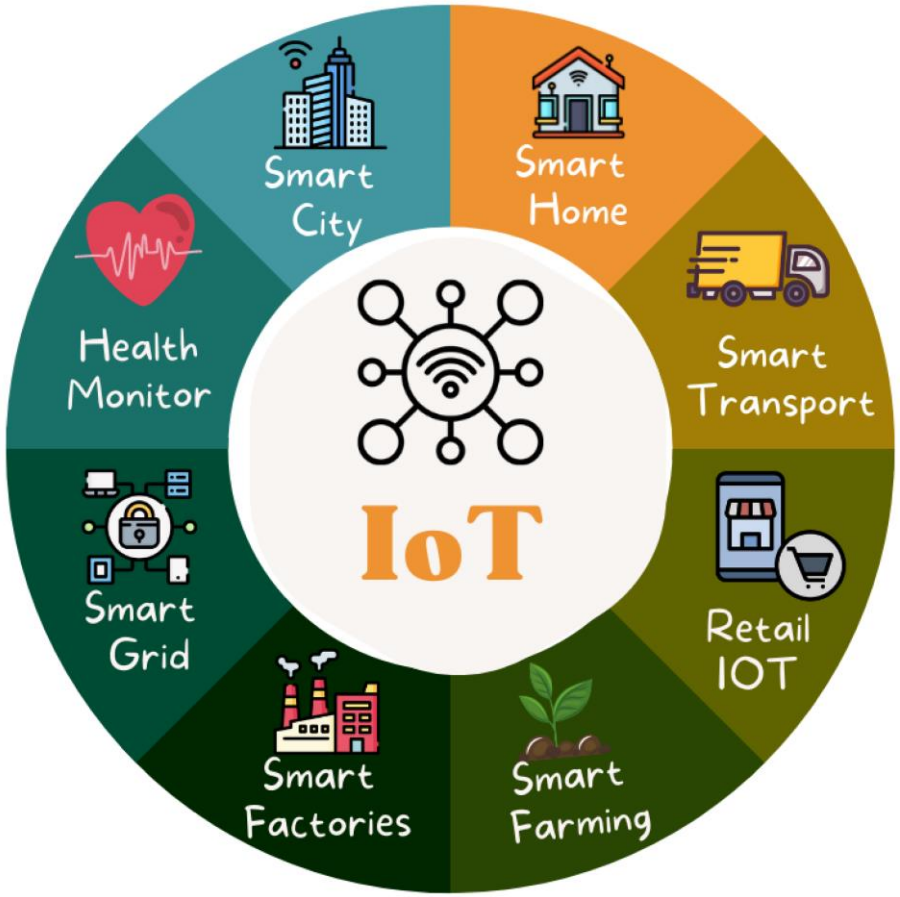
Introduction to IoT

The **Internet of Things (IoT)** is a vast ecosystem of **interconnected physical devices** (sensors, actuators, machines, vehicles, wearables, etc.) that **collect, exchange, and act on data** through the Internet.

IoT bridges the gap between the **physical world** and the **digital world**, enabling devices to communicate autonomously and make intelligent decisions without human intervention.

Key Characteristics:

- Connectivity
- Intelligence (data-driven decision-making)
- Interoperability
- Scalability
- Automation

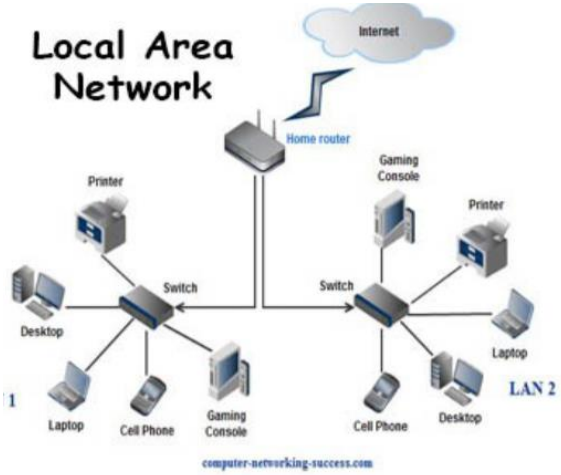


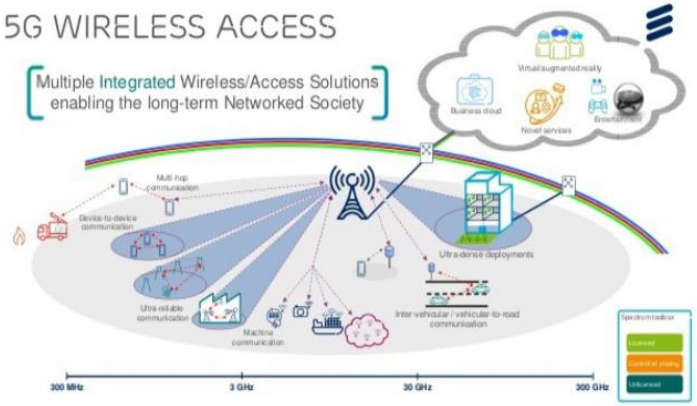
IoT - Historical Perspective

Yesterday
local Area Networks (LAN), Staticservices (email, web)

Today
Wireless Mobile, ubiqitousInternet access, Cloud MobileServices & Resources

Today/Tomorrow
Machine to machine (M2M), Internet ofThings (IoT), Smart World services





Evolution

Evolution and Motivation for IoT

Historical Progression:

- **M2M (Machine-to-Machine)** communication (pre-2000s) – closed industrial networks.
- **IoT Emergence (2010–2020):** Cloud, mobile networks, and embedded systems integration.
- **Next-Gen IoT (2020–present):** Driven by **5G**, **AI**, and **Edge Computing**.

Motivations:

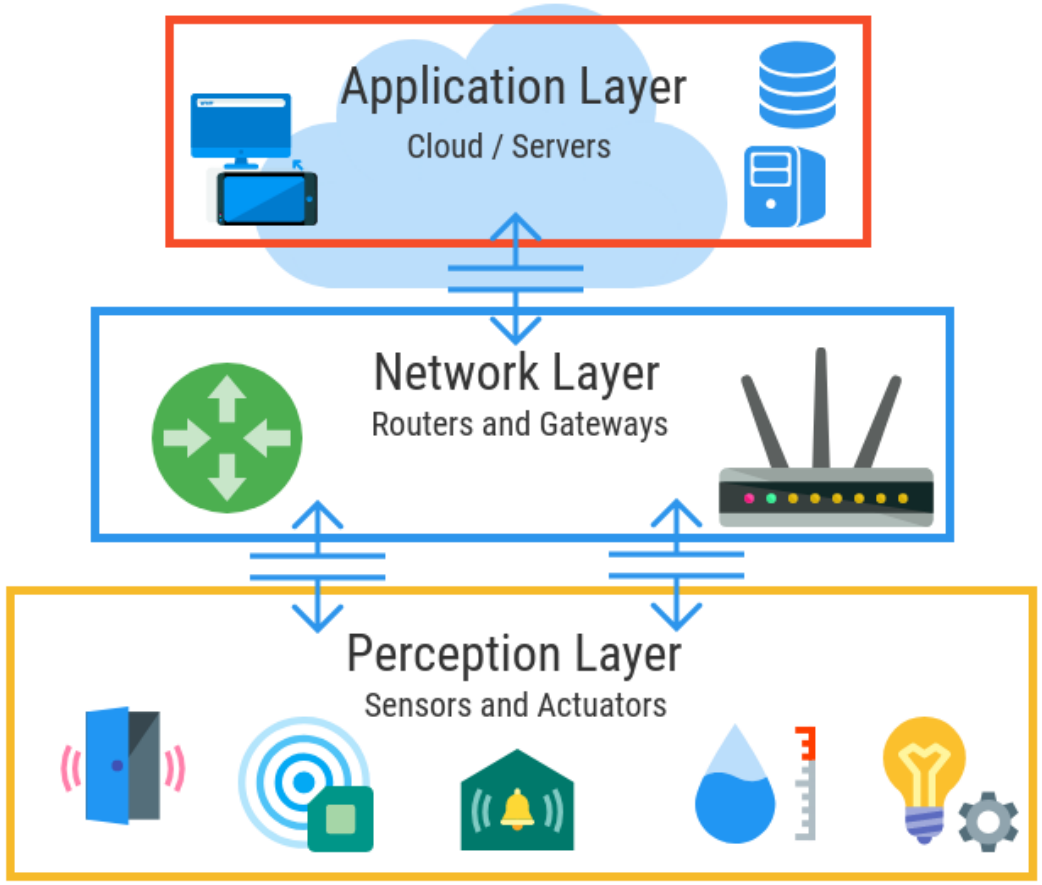
- Need for **automation** and **real-time decision-making**.
- Massive availability of **low-cost microcontrollers** and **sensors**.
- Growth of **wireless connectivity** (Wi-Fi, LPWAN, 5G).
- Transition to **data-driven ecosystems** and **smart services**.

Statistics (2025 projection):

30+ billion connected IoT devices worldwide.

Over 60% of industrial operations use IoT-based analytics.

IoT 3-Layer Architecture

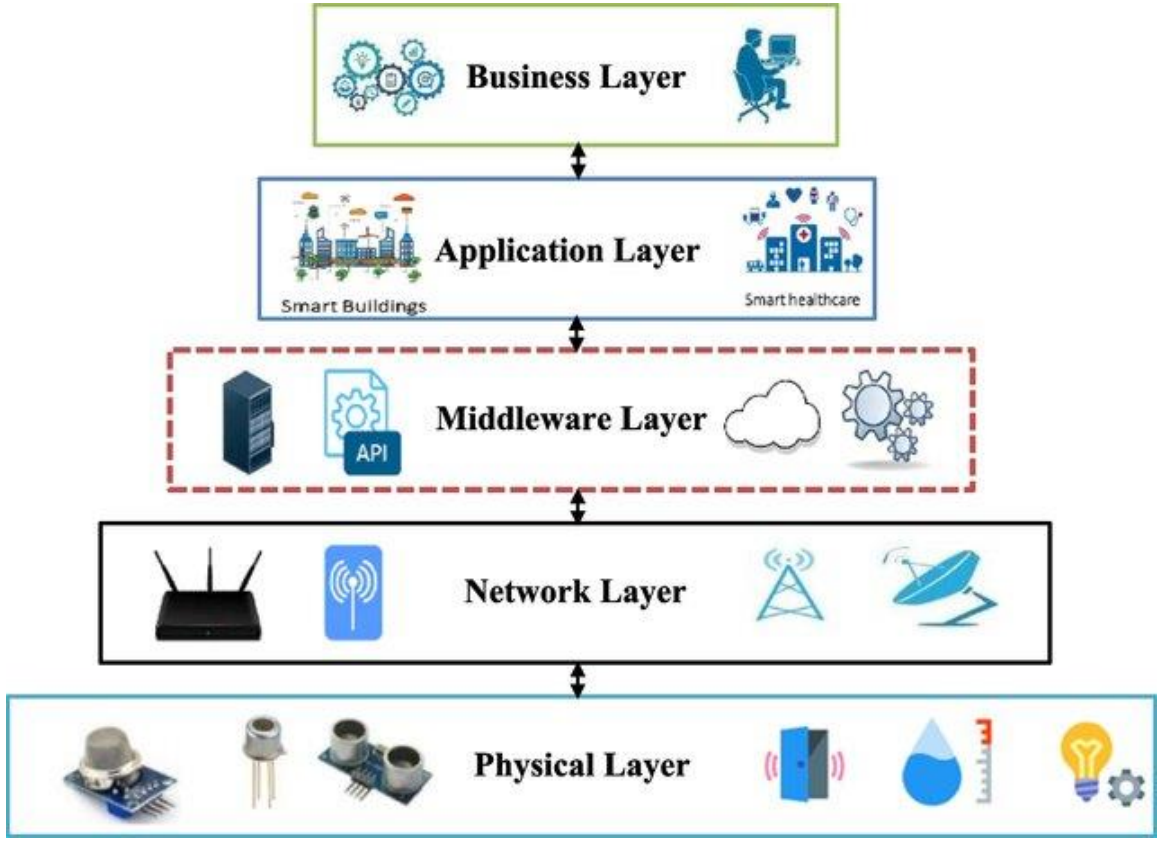


Service delivery and user interaction

Data transmission and routing between devices/cloud

Data acquisition and environmental sensing

Advanced 5-Layer IoT Architecture



Business Layer – Manages business logic, user policies, billing, and overall IoT system integration.

Application Layer – Provides domain-specific services (smart home, industry, healthcare).

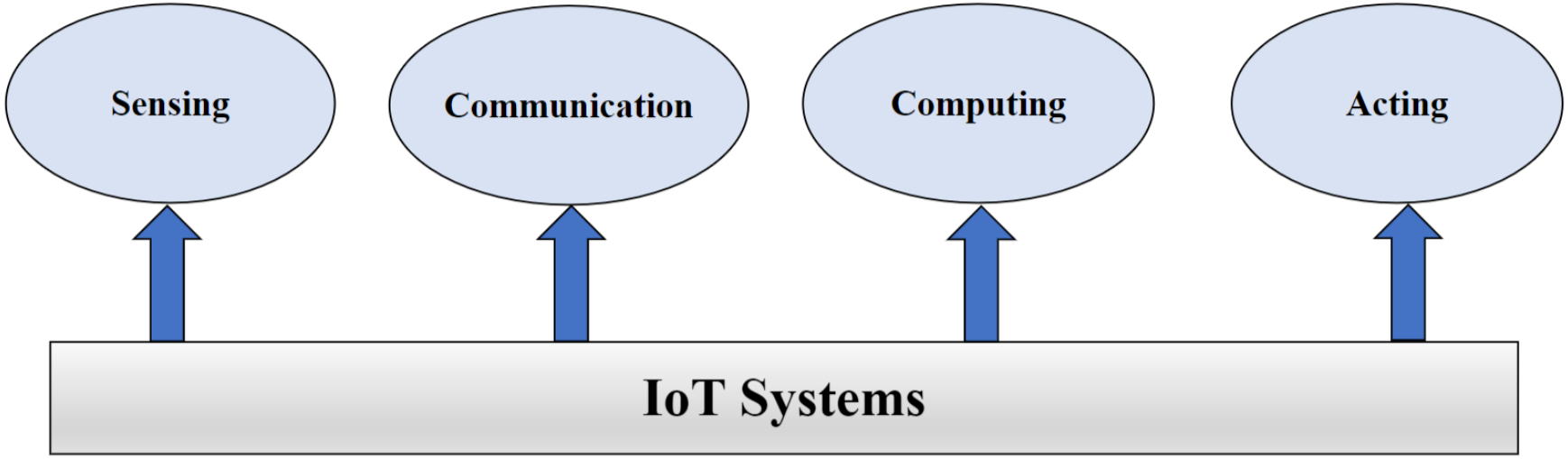
Processing (Middleware) Layer – Performs data aggregation, filtering, analytics, and storage (Cloud or Edge).

Network Layer – Transmits data securely using IPv6, 6LoWPAN, LTE-M, or NB-IoT.

Perception Layer – Captures data from the physical world (e.g., temperature, humidity).

Four Main Components of IoT Systems

- Sensor: a connected device enabling the sensing of physical parameters of the scenario or controlled environment, those values are transformed into digital data.
- Actuator: a connected device enabling the activation of actions on the controlled environment.
- Controller: a connected device implementing an algorithm to transform input data in actions.
- Smart things: digital devices providing service functions realized by the synergy between sensors, actuators and controllers (possibly implemented by local/distributed execution platforms and M2M/Internet communications).



IoT Communication Models

› IoT devices communicate in several topologies or interaction models:

Model	Description	Example
Device-to-Device (D2D)	Direct link via Bluetooth, ZigBee	Smart lock ↔ motion sensor
Device-to-Gateway	Local gateway aggregates and forwards data	Smart home hub to cloud
Device-to-Cloud	Direct connection to cloud servers	Wi-Fi camera to AWS IoT Core
Device-to-Edge	Local computation at edge nodes before cloud	Factory edge server processing sensor data
Device-to-Mobile	Smartphone acts as a control interface	Smartwatch linked to mobile health app

IoT Communication Technologies

Technology	Range	Data Rate	Power	Frequency	Use Case
Wi-Fi (802.11n/ac/ax)	100 m	Up to 1 Gbps	Medium	2.4/5 GHz	Smart homes
Bluetooth/BLE	10–30 m	1–3 Mbps	Low	2.4 GHz	Wearables, accessories
ZigBee (802.15.4)	10–100 m	250 Kbps	Very Low	2.4 GHz	Home automation
LoRa/LoRaWAN	2–15 km	<50 Kbps	Very Low	Sub-GHz	Smart cities, agriculture
NB-IoT / LTE-M	Several km	100–1000 Kbps	Low	Licensed LTE bands	Industrial IoT
5G mMTC	Large	Up to 10 Mbps	Low–Medium	Sub-6GHz, mmWave	Massive IoT connectivity

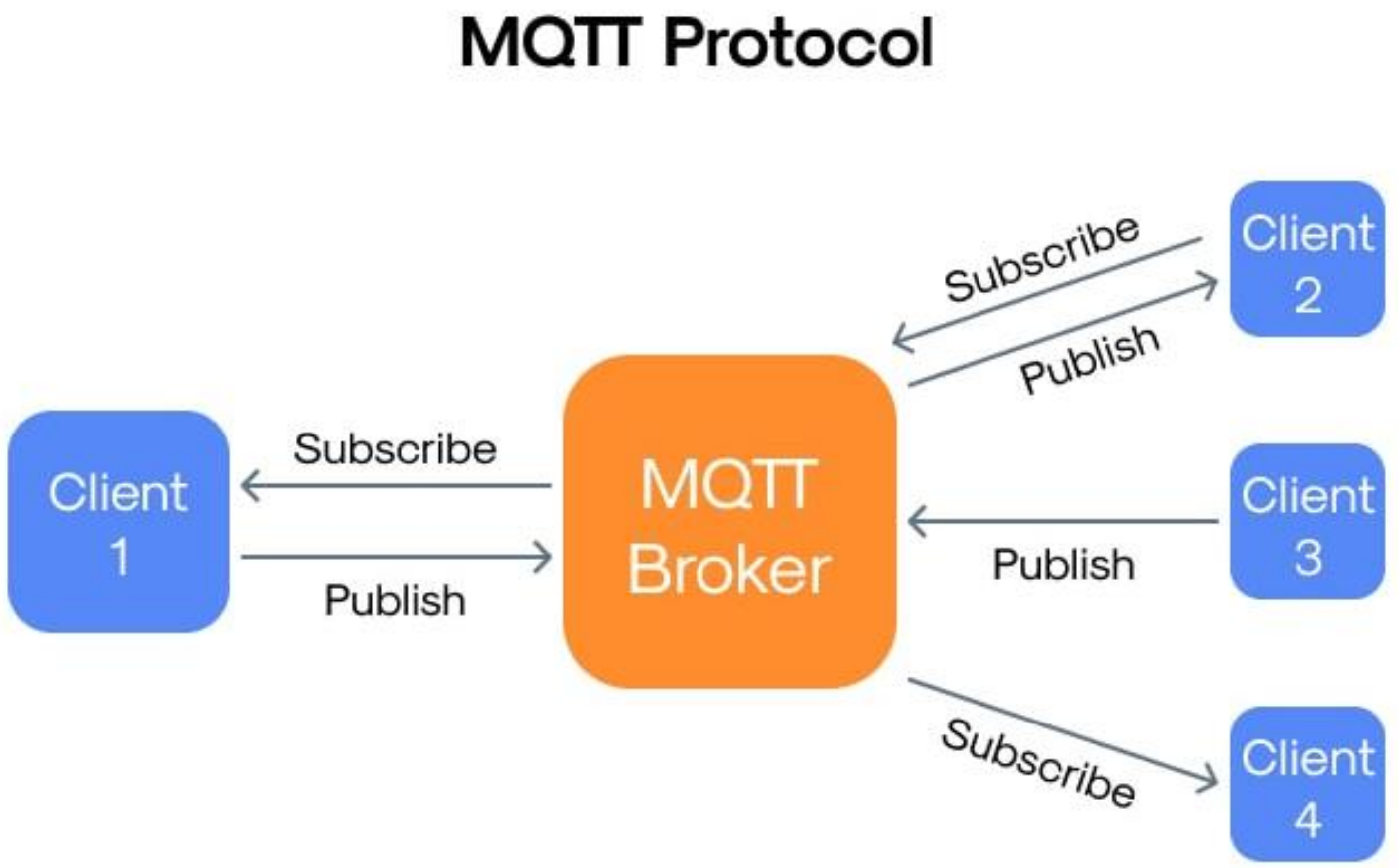
IoT Cloud

- The **IoT Cloud** is a **distributed cloud platform** designed specifically to collect, store, process, and analyze data generated by IoT devices.
- It serves as the **middleware** and **application backbone** of modern IoT systems, connecting the physical world (sensors, actuators) with cloud-based applications and analytics.

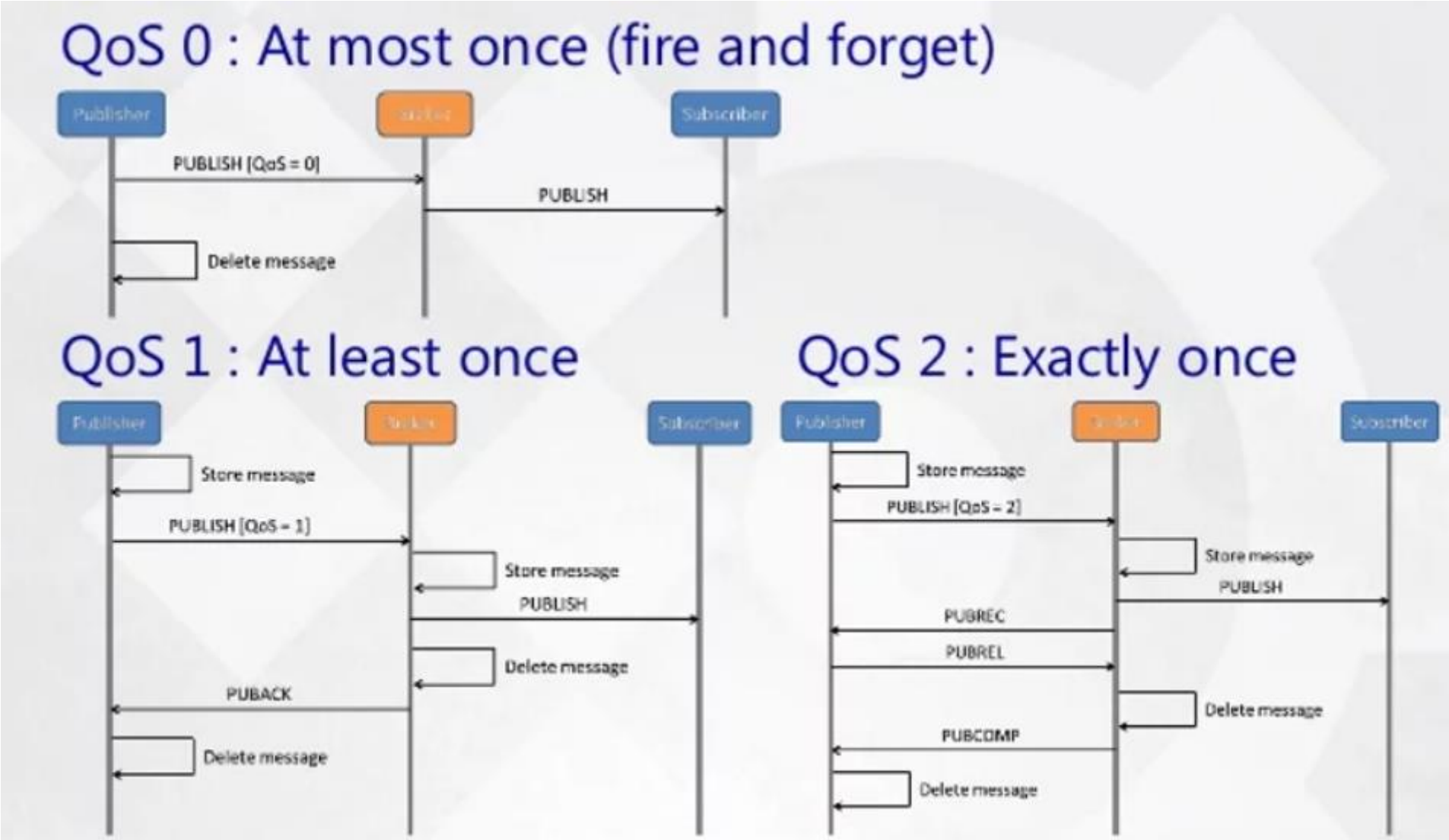
Data Collection	Gathers sensor data via gateways or directly from devices using protocols like MQTT, CoAP, or HTTP.
Data Storage	Stores massive streams of time-series data in scalable databases (SQL, NoSQL, InfluxDB, etc.).
Processing & Analytics	Uses real-time analytics, machine learning, and AI to extract insights and trigger actions.
Device Management	Registers, monitors, updates, and controls IoT devices remotely.
Integration	Exposes APIs for other systems or applications (mobile apps, dashboards, enterprise systems).

MQTT Protocol (Application Protocol)

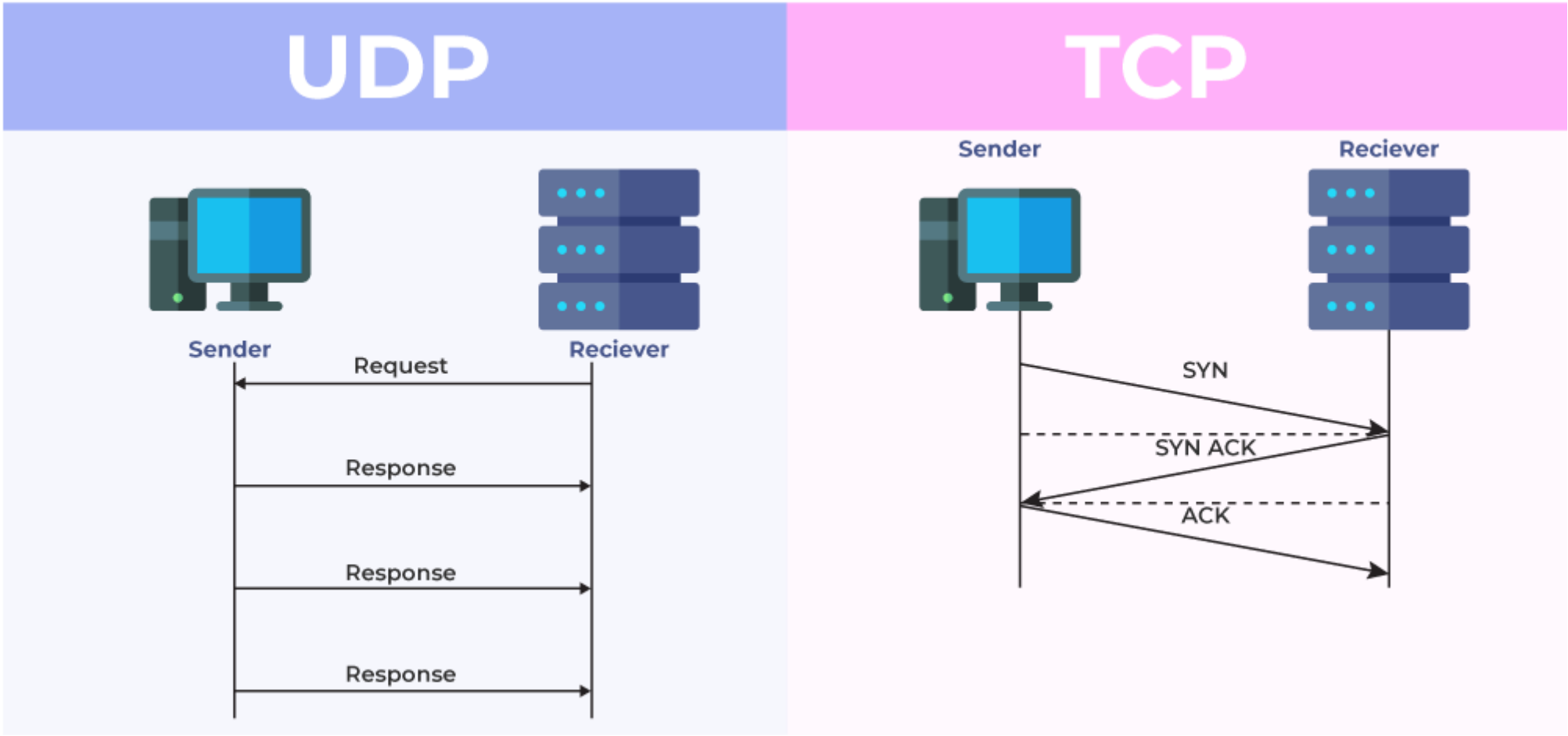
MQTT (Message Queuing Telemetry Transport) is a **lightweight, publish/subscribe** messaging protocol designed for **low-bandwidth**, **high-latency**, or **unstable networks** — exactly what IoT devices often face. It runs over **TCP/IP** and ensures **reliable communication** between constrained devices and cloud services.



MQTT QoS


QoS 0 : At most once (fire and forget)
Publisher
Subscriber
PUBLISH [QoS = 0]
PUBLISH
Delete message
QoS 1 : At least once
Store message
PUBLISH [QoS = 1]
Store message
PUBLISH
Delete message
PUBACK
Delete message
QoS 2 : Exactly once
Publisher
Subscriber
Store message
PUBLISH [QoS = 2]
Store message
PUBLISH
PUBREC
PUBREL
Delete message
PUBCOMP
Delete message

Network Layer protocols



Network Layer protocols (IPv6)

Why IPv6?

- IoT networks may include **billions of devices**, each needing a unique address. IPv4 (with ~4.3 billion addresses) is insufficient, while **IPv6 provides 3.4×10^{38} possible addresses**, easily accommodating IoT's massive scale.

- IPv6 enables **direct device-to-device communication**, removing the need for NAT (Network Address Translation), simplifying routing, and improving efficiency.

- fe80::1ff:fe23:4567:890a

Network Layer protocols (IPv6)

Feature	Description	Benefit for IoT
Larger Address Space	128-bit addresses (vs 32-bit in IPv4).	Each IoT device can have a unique, globally routable address.
Simplified Header Format	Fewer fields → faster processing.	Lower latency and better performance for constrained devices.
Auto-configuration (SLAAC)	Devices can self-configure addresses without DHCP.	Simplifies deployment of thousands of IoT sensors.
Built-in Security (IPsec)	Mandatory in IPv6 specifications.	Ensures data integrity and authentication in IoT communications.
No NAT Needed	Direct addressing between devices.	Enables true end-to-end connectivity and easier management.
Better Multicast Support	Efficient group communication.	Reduces bandwidth for sensor networks sending periodic updates.
Mobility Support (MIPv6)	Native support for mobile devices.	Important for mobile IoT (vehicles, wearables, drones).

IoT Security Challenges

Aspect	Vulnerability	Example Attack	Countermeasure
Device Layer	Weak passwords, unpatched firmware	Botnets (Mirai)	Strong credentials, firmware updates
Network Layer	Unencrypted data, spoofing	Packet sniffing	VPNs, DTLS, TLS
Application Layer	Poor authentication	Unauthorized access	OAuth 2.0, Token-based auth
Data Privacy	Sensitive data leakage	Cloud compromise	Encryption, access control
Scalability	Device management issues	Configuration drift	Centralized IoT management